

Barcelona School of Economics

Master in Specialized Economic Analysis
International Trade, Finance and Development

"Climate Shocks on Chilean Copper Mining: ENSO-attribution, Impact, and Adaptation"

Authors:
Gleb Enson Rohan Kale Manuel Seick Julia Suárez Fatimah Yasin

Directors:
Professor Fernando Broner
Professor Manual Garcia Santana

9 June 2026

ABSTRACT IN ENGLISH (100 words):

Chile is the world's leading copper producer, repeatedly exposed to climate shocks linked partly to ENSO. We build a mine-level panel for 19 mines, separating specific operational-threshold crossings from their ENSO-attributable subset. The results are channel-specific. Unusually wet months lower output by a robust 4.7%; the ENSO-attributable flood subset is negative (2.5%), unresolved. Heat shows no response. Drought is a slow channel: general exposure is null, ENSO-attributed drought lowers annual output in 3-year window, coincide with the late megadrought. The panel cannot estimate adaptation causally; it is illustrated through the muted output response, water-rights management, and firms' investment in desalination.

ABSTRACT IN SPANISH (100 words):

Chile es el principal productor mundial de cobre y está expuesto de manera recurrente a perturbaciones climáticas vinculadas en parte al ENSO. Construimos un panel a nivel de mina para 19 operaciones, separando los cruces de umbrales operacionales específicos de su subconjunto atribuible al ENSO. Los resultados son específicos por canal. Los meses inusualmente húmedos reducen la producción en un 4,7%, un efecto robusto; el subconjunto de inundaciones atribuible al ENOS es negativo (2,5%), aunque no estadísticamente resuelto. El calor no muestra respuesta. La sequía es un canal lento: la exposición general es nula, mientras que la sequía atribuida al ENOS reduce la producción anual en una ventana de tres años, coincidiendo con la megasequía tardía. El panel no permite estimar la adaptación de forma causal; ésta se ilustra a través de la respuesta amortiguada de la producción, la gestión de los derechos de aprovechamiento de agua y la inversión de las empresas en desalación.

KEYWORDS IN ENGLISH (3)**:** CLIMATE ADAPTATION; COPPER MINING; EL NIÑO–SOUTHERN OSCILLATION (ENSO)

KEYWORDS IN SPANISH (3)**:** ADAPTACIÓN CLIMÁTICA; MINERÍA DEL COBRE; EL NIÑO–OSCILACIÓN DEL SUR (ENOS)

Contents

1 Introduction **1**

2 Literature Review **2**

3 Data **3**
- 3.1 Output and Climate Data . . . 3
- 3.2 Desalination and the Adaptation Variable . . . 5
- 3.3 Firm-Level and Institutional Sources . . . 5

4 Methodology **6**
- 4.1 Climate Shocks Model: Shock Definition, ENSO-attribution and Empirical Design . . . 6
- 4.2 Desal-Adaptation Model: Desalination and Output . . . 9

5 Results **10**
- 5.1 Event Accounting . . . 10
- 5.2 Impact of Climate Shocks on Copper Mining Output . . . 12
- 5.3 Testing the Adaptation Margin . . . 13

6 Adaptation in Practice **15**
- 6.1 Anticipatory responses to ENSO forecasts . . . 15
- 6.2 Institutional and Water-Governance Response . . . 17
- 6.3 Footprint of Water Stress: Costs, Provisions, and the Balance Sheet . . . 19

7 Limitations **21**

8 Conclusion **22**

A Robustness Checks **26**
- A.1 Climate Shocks Model . . . 26
- A.2 Desal-Adaptation Model . . . 28

B Additional Results and Figures **32**
- B.1 Data . . . 32
- B.2 Climate Anomalies and Event Accounting . . . 33
- B.3 Output Impact of Climate Shocks . . . 35
- B.4 Firm-Level Response and ENSO Forecasts . . . 36
- B.5 Institutional and Water-Governance Response . . . 38
- B.6 Financial Evidence . . . 40

1 Introduction

Economists conventionally measure the cost of climate stress as lost output: when conditions turn anomalous — too hot, too dry, or too wet — production decreases (Dell et al. 2014). Such conditions may be connected to global climate fluctuations. One of the most studied drivers of climate anomalies is the El Niño–Southern Oscillation (ENSO) — a recurring fluctuation in sea-surface temperatures and atmospheric pressure across the equatorial Pacific that cycles irregularly every two to seven years between a warm phase (El Niño), a cool phase (La Niña), and a neutral state, swinging exposed regions between wetter and drier extremes (NOAA Climate Prediction Center 2024). Few places are as exposed to those fluctuations as Chile, the longest country in the world from north to south: a four-thousand-kilometer strip pinned between the Andes and the Pacific that runs from the hyper-arid Atacama to a rain-fed centre. The strongest ENSO episodes — 1982/83, 1997/98, 2015/16, 2023/24 —left their mark on Chilean weather records, and the notorious megadrought, beginning in 2010, was the most severe in the country's documented history (Garreaud et al. 2017).

The recurring nature of ENSO and the extremely diverse geography of Chile make research on adaptation patterns in various sectors of Chilean economics gripping. We study this recurring shock in copper mining. The choice is not exclusively one of economic weight, though that weight is considerable — mining is roughly a tenth of Chilean GDP and accounts for well over half of export share (Comisión Chilena del Cobre (COCHILCO) (2024)); Chile, being a home for the largest copper mine in the world — Escondida — alone supplies about a quarter of the world's mined copper (U.S. Geological Survey (2024)). Moreover, it is grounded in that, almost uniquely among economic activities, a mine cannot choose the climate in which it operates. A farm or factory can be re-sited, but a mine is built on its orebody — and Chile's orebodies lie in the arid north, fixed there by a geology indifferent to the weather above. The country's most valuable capital is therefore immobile, sunk for decades in precisely the regions where ENSO's drought swings bite hardest. This is what makes copper the natural place to analyze the cost of a recurring climate shock: the hazard returns cyclically, the asset cannot move away from it, and the only margin left is to adapt in place.

These features raises a question: does the Chilean copper mining sector react differently to climate shocks induced by ENSO? The question is broad, so we break it into three more refined probes that the paper takes in turn. First, *attribution*: are the climate shocks reaching Chile's copper mines genuinely ENSO-driven, or merely the background extremes of an already harsh environment? Second, *impact*: how much do those shocks reduce copper output, and through which channels? Third, *response*: how does the industry adjust?

To answer these questions we build a panel of Chilean mines observed monthly and develop a methodology for climate shocks, based on relevant operational thresholds (section 4.1). We isolate the ENSO attributable share with a separate flag, that fires only when an active ENSO-phase, the region's historical teleconnection, the realised local anomaly, and an operational threshold coincide (results in section 5.1 and section 5.2). We then attempt to empirically confirm an adaptation for the most strategically important shock — water-scarcity (section 4.2 and section 5.3). Finally, we explore the application of forecastability of ENSO (section 6.1), dive into institutional response (section 6.2) and investigate footprints of climate shocks in the financial statements of the firms (section 6.3). The rest of the paper is organized as follows: section 2 reviews the related literature; section 3 describes the data and section 8 concludes our findings and describes the limitations we faced.

2 Literature Review

This paper draws on four literatures: the panel-based measurement of climate impacts, the macroeconomics of the El Niño–Southern Oscillation (ENSO), the transmission of weather shocks through trade and firm performance, and the economics of adaptation. We review each in turn and state where our study departs from existing work.

The new climate–economy literature. Modern empirical work identifies the effect of the climate on the economy from year-to-year weather variation within fixed spatial units, using panel fixed effects to absorb time-invariant confounders (Dell et al. 2014; Kolstad and Moore 2018). This "weather" approach is now the standard alternative to the cross-sectional Ricardian method of Mendelsohn et al. (1994), which infers long-run impacts from how outcomes vary across climates but is exposed to omitted-variable bias. We adopt the panel approach, but apply it at the level of the individual mine cell rather than the country or region, and we replace the conventional continuous temperature and precipitation regressors with an explicit, engineering-motivated definition of operational climate stress.

Defining and attributing ENSO. A growing macro-finance literature studies ENSO specifically, measuring it from sea-surface-temperature anomalies in the equatorial Pacific and relating it to output (Liu et al. 2023), sovereign and banking risk (Damette et al. 2024, 2025), and commodity and asset prices (Atems and Sardar 2021; Ubilava 2017). Identification strategies range from standard-deviation event dummies (Liu et al. 2023) to data-driven regime classifications (Dufrénot et al. 2024) and smooth-transition functions (Ubilava 2017). A common feature is that ENSO enters as a single global signal applied uniformly across the panel—one annual dummy or regime per year, identical across space. Our contribution to this strand is to localize and attribute the shock: we require that an active ENSO phase, the historical regional teleconnection, and the realized local anomaly all align before a mine-month is counted as ENSO-attributable stress, separating ENSO-driven disruption from generic threshold crossings.

Climate shocks, exports, and firm performance. Because copper is Chile's dominant export, our work also connects to research on how climate shocks travel through trade and firms. Weather shocks reduce exports at the country level (Jones and Olken 2010) and at the firm level in developing economies (Nedoncelle and Wolfersberger 2025), depress growth (Cevik and Jalles 2023), and shape firm outcomes conditional on supply-chain resilience (Luo et al. 2026). This work generally treats output or exports as the outcome and reads damage off their decline. We instead study a single export-critical sector at the level of the producing asset and, as the next strand motivates, do not presume that the damage must surface in output at all.

Adaptation. The most directly related literature concerns adaptation, where a recurrent finding is that estimated damages shrink once agents are allowed to adjust: the United States temperature–mortality relationship fell sharply over the twentieth century as air conditioning diffused (Barreca et al. 2016), and short-run agricultural impacts exceed those that survive medium-run adaptation (Burke and Emerick 2016), while the Ricardian tradition embeds adaptation by construction (Mendelsohn et al. 1994). The methodological

lesson—that an attenuated or null output response can reflect successful adaptation rather than the absence of a shock—is central to our paper. Where this literature typically infers adaptation *indirectly*, from the attenuation of estimated damages across time or space, we observe the adaptation margin *directly*, in the Chilean copper sector's structural shift to desalinated seawater. Our central argument is that the muted output response to water stress reflects impact absorbed on the capital and cost margins, not the absence of impact—and we document that absorption rather than inferring it.

Mining and the Chilean setting. Finally, a sector- and country-specific literature studies climate risk in mining and in Chile. Work on mining and climate is dominated by forward-looking risk assessment: projection-based, mine-by-mine exposure mapping (Odell et al. 2018), composite multi-hazard indices linked to firm financial performance (Sun et al. 2020), and integrated hazard–exposure–vulnerability scores for Chilean mining regions (Del Rio et al. 2023). For Chile, Hernández and Madeira (2021) estimate climate effects on regional output in a region–industry panel, and Beltrán et al. (2023) provide a macroeconomic assessment. Relative to this work, we provide, to our knowledge, the first ex-post, mine-level, ENSO-attributed causal panel for Chilean copper output, paired with direct evidence on the adaptation mechanism—complementing the institutional and water-governance literature with production- and firm-level evidence.

Across these strands, our common departure is that we attribute operational climate stress to ENSO at the individual mine level, and we read the sector's response off the adaptation margin directly—firm accounts and the structural shift to desalinated seawater—rather than off the decline in output alone.

Contribution. Our central contribution is methodological: a mine-cell design that isolates the ENSO-attributable share of operational climate stress from the background aridity of an already harsh environment, and lets us measure its output cost channel by channel. As that cost turns out to be small, the second contribution is to ask where the climate's bite actually surfaces, and to answer it with firm- and institutional-level evidence rather than the output panel: on operating margins, capital expenditure, and balance sheets—a margin standard output-based damage estimates cannot see. We document this absorption rather than estimate it, and we are explicit throughout about the line between the two.

3 Data

3.1 Output and Climate Data

Mine output and price. The dependent variable is monthly fine-copper output (in kilotonnes) from the COCHILCO monthly archive, which covers the nineteen operations that together account for roughly 95% of national mine output. Output is available from January 2003 and enters in logs. Zero-output mine-months—pre-commercial ramp-ups and whole-mine standstills (Codelco-Salvador's 2024–25 Rajo Inca rebuild; Cerro Colorado's 2023–25 closure)—are non-climate events and are dropped. The within-year price control is the LME monthly copper price (US dollars per tonne, logged); because that series begins in 2006, the estimation sample is the priced, active-mine-month panel of 4,393 mine-months over 2006–2025, with a parallel 369 mine-year panel for the slow drought channel.

Geography and the mine-to-cell match. The nineteen operations span the seven copper-bearing regions—Tarapacá, Antofagasta, Atacama, Coquimbo, Valparaíso, O'Higgins, and the Santiago Metropolitan Region. Each mine is matched to the 0.05° climate grid cell containing its operating coordinates; regional fields are area-weighted over the INE 2017 region polygons. The mines divide sharply by altitude—high-cordillera operations ($\geq 2{,}000$ m) versus low-altitude pits—a split that governs whether a documented regional flood actually binds at a given site (Appendix B).

Climate side. The climate side runs 1960–2025 and is built from CR2MET v2.5 (daily, 1960–2021) extended with ERA5-Land (2022–2025). Bias correction is panel-specific: the acute daily fields (precipitation and maximum temperature, for flood and heat) are quantile-mapped at the mine cell against the 2017–2021 CR2MET distribution within each (mine, calendar-month); the twelve-month SPEI is computed from monthly ERA5-Land temperature and precipitation, bias-corrected per (mine, calendar-month) on the 2012–2021 overlap. ENSO state is the NOAA three-month rolling ONI; the regional teleconnection ρ and standardized anomaly Z are computed on the 1980–2010 baseline (see the methodology section). The three windows play distinct roles: 1980–2010 is the stable, pre-megadrought normal that defines the standardized anomaly Z and the ENSO teleconnection ρ, while the 2017–2021 (daily acute fields) and 2012–2021 (monthly SPEI) windows are bias-correction overlaps that harmonize the products onto the recent CR2MET scale and do not enter the anomaly definition.

Financial data. The panel covers fourteen operations over 2006–2025, constructed from audited IFRS statements filed with the CMF or the relevant home regulator. Six variables are extracted—revenue, net income, current and non-current liabilities, capital expenditure, and closure provision—normalized throughout to USD millions. C1 cash cost (US cents per pound, net of by-product credits) is drawn from parent-company operating reviews: BHP Annual Reports for Escondida and Spence, Anglo American Annual Reports for Collahuasi and Anglo American Sur, and Codelco's *Principal Información Financiera*; four operations carry no publicly reported mine-level C1. Coverage gaps reflect disclosure practice and are preserved rather than imputed. Month-level stress flags are aggregated to calendar-year indicators before merging: a mine-year receives a drought flag if at least one constituent month satisfies the attribution criterion.

The master panel. All primitives—the physical observation, the operational threshold, the three ENSO components (ONI, ρ, Z), and the resulting stress flag—are joined into a single auditable master panel with one row per mine $\times$ month $\times$ channel ($19 \times 792 \times 5 = 75{,}240$ rows), from which every flag is re-derivable by inspection. Copper output is strongly right-skewed (a median of 12.4 against a mean of 22.7 kt, reflecting the gap between the Escondida-scale majors and the small Atacama pits); the ONI spans a full La Niña–El Niño range over the window (-1.6 to $+2.8$); and the ENSO-attributable stress flags fire on only a small minority of mine-months.

3.2 Desalination and the Adaptation Variable

The desalination registry. To measure adaptation we assemble a mine-level desalination registry for the nineteen panel operations, recording whether and when each mine's dominant production process moved onto desalinated (or direct) seawater. Coding is built from three layers: operator annual reports filed with the Comisión para el Mercado Financiero (CMF *Memorias*), which disclose water sources; COCHILCO's public *Consumo de Agua en la Minería del Cobre* series; and the ACADES desalination registry, which records commissioning dates and plant capacities. The headline variable, $\text{Adapt}_{i,t}$, is the resulting indicator used in eq. (7); its construction and variants are described in section 4.2.

3.3 Firm-Level and Institutional Sources

Beyond the output and climate panel, the evidence on adaptation in section 6 draws on three further public sources, each supporting one strand of that chapter.

ENSO forecasts and permit filings. The test for anticipatory adaptation draws on two records. The forecast record is the monthly consensus probabilistic ENSO outlook issued jointly by the NOAA Climate Prediction Center and the IRI, publicly archived, from which we read the probability assigned to each phase for each forthcoming three-month season. Environmental-permit activity is taken from the public e-SEIA (*Sistema de Evaluación de Impacto Ambiental*) filing database, from which we date water- and flood-related infrastructure submissions for the nineteen operations and test whether filings cluster in the forecast window.

Water-governance data. Spatial information on Chile's water-governance constraints comes from the Dirección General de Aguas (DGA)—its Catastro Público de Aguas and associated Mapoteca Digital (updated November 2025)—which provide georeferenced layers for aquifer restriction and prohibition zones, exhausted-river declarations, drought-scarcity decrees, protected wetlands and glaciers, and Article 5 reserved flows. We combine these layers with the georeferenced locations of the nineteen operations, drawn from the project mine registry, to situate each mine relative to legally constrained water bodies. Sector-level water use is from COCHILCO's annual *Consumo de Agua en la Minería del Cobre* (2024), which records mining withdrawals by source—surface, groundwater, third-party, and seawater.

Firm-level financial statements. The cost and balance-sheet evidence in section 6 is taken from operator financial statements—annual reports (*Memorias Anuales*) filed with the CMF—from which we extract C1 cash cost, capital expenditure, revenue, and IAS 37 closure provisions. Coverage is uneven across operators and years: not every mine reports a continuous cash-cost series, so this evidence is descriptive rather than a balanced panel, and we report each series for the operators and years in which it is disclosed. Statements where available are consolidated at a parent level with several mines combined in a single statement. Firm-level water-consumption microdata (COCHILCO's EMPAE) remain confidential, as noted in section 3.2.

4 Methodology

4.1 Climate Shocks Model: Shock Definition, ENSO-attribution and Empirical Design

Channels for the climate shocks and mines' response. To define the empirical climate shock flags, it is necessary to identify the channels through which climate shocks affect physical copper output. From the mine engineering and climate-impact literature, three channels are derived: flood, heat and drought.

Each has specific effects and a set of mine response measures. Flood and heat are *acute operating hazard channels*. Intense precipitation can flood pits and haul roads, wash out access routes and ports, and raise tailings-water risks; the feasible response is mainly operational, through early-warning systems, stormwater protocols, temporary stoppages, tailings security, and rerouting of concentrate logistics. Heat impacts through worker stress, excessive equipment wear, dust, and evaporation: in response, mines can shift work timing, enforce DS 594 heat-stress protocols, increase hydration and dust suppression, and invest in equipment cooling and ventilation.

Drought is a distinct, *slow operating hazard channel* of strategic importance. Water is a direct production input in copper concentration. Reacting to water scarcity in the short run, mines can reduce throughput, prioritize high-grade ore, recycle more process water, truck in water in exceptional cases—or even completely stop production if water rights are restricted. In the long run, the relevant strategic response is capital-intensive water substitution: investments in building desalinated or direct seawater systems, pipelines, recycling infrastructure, and new water-rights arrangements.

These differences motivate the empirical split below: floods and heat are estimated as acute mine-month shocks, while drought is treated as an accumulated water-stress channel.

Definition of climate shock. The climate shocks model in our work asks whether mine output falls when climate conditions cross an operationally meaningful threshold, defined for each of the mentioned channels, and whether the threshold-crossing month is consistent with the contemporaneous ENSO teleconnection. The design therefore separates two objects. The first is a *general* climate shock: any physical threshold crossing, regardless of whether it is ENSO-attributable or not. The second is an *ENSO-attributable* shock: the subset of threshold crossings that also occurs during an active ONI phase, in the direction predicted by the regional teleconnection, and with the realized local anomaly moving in that same direction. Thus the results should be read as a contrast between the impact of all general climate shocks and that of the ENSO-attributed subset.

Table 1. Operational thresholds and their roles in the climate shock model. The regional-mean monthly accumulation is the headline flood definition; the mine-cell daily variant is a conservative point-level lower bound.

Channel	Exposure	Operational threshold	Role in the analysis
Flood, regional mean[1]	Area-weighted regional precipitation in month m	$P_{r,m} \geq 20$ mm/month	Headline acute flood measure; captures broad wet-month disruption.
Flood, mine cell[2]	Mine's own 0.05° grid cell, daily precipitation	Any day $P_{i,d} \geq 20$ mm	Conservative point-level lower bound.
Heat, mine cell[3]	Monthly mean mine-cell maximum temperature	$T^{\max}_{i,m} \geq 23.5\,^{\circ}\mathrm{C}$	Routine heat-exposure channel at monthly frequency.
Drought, mine cell[4]	Mine-cell SPEI-12, annualized into lagged exposure shares	SPEI-12 ≤ -1	Slow water-stress channel.

The flood and heat thresholds are evaluated at mine-month frequency. The drought threshold is first observed monthly and then accumulated into annual mine-year exposure measures because water scarcity is a stock rather than a one-month shock.

[1] Wilcox et al. (2016) 20 mm wet-event threshold, applied at the regional-mean monthly scale for the headline flood measure.

[2] Wilcox et al. (2016) alluvion threshold, applied as a mine-cell daily point-exposure lower bound.

[3] Chile DS 594/1999 occupational heat-stress rule; the daily 27 °C anchor is calibrated to a CR2MET monthly-mean threshold.

[4] WMO moderate-drought convention using SPEI-12, following the standardized water-balance index of Vicente-Serrano et al. (2010).

ENSO-attribution: anomalies and teleconnection primitives. For each climate variable v and region r, the realized climate value is standardized against the 1980–2010 calendar-month baseline,

$$Z^{v}_{r,m,t} = \frac{X^{v}_{r,m,t} - \mu^{v}_{r,m}}{\sigma^{v}_{r,m}}, \tag{1}$$

where $X^{v}_{r,m,t}$ is the observed precipitation or temperature value, $\mu^{v}_{r,m}$ is the baseline mean, and $\sigma^{v}_{r,m}$ is the baseline standard deviation. The baseline period is chosen to fall in the pre-megadrought window, where climate-data quality is higher than in earlier periods. The regional teleconnection $\rho^{v}_{r,m}$ is the 1980–2010 Pearson correlation between the same regional climate variable and the Oceanic Niño Index in calendar month m. It supplies the expected direction of the local anomaly during El Niño or La Niña.

The ENSO-consistency indicator for channel c is the product of three conditions:

$$\mathrm{ENSOConsistent}^{c}_{r,y,m} = \underbrace{\mathbf{1}\left[\,|\mathrm{ONI}_{y,m}| \geq 0.5\right]}_{\text{(i) active ENSO phase}} \cdot \underbrace{\mathbf{1}\left[\,\mathrm{sign}\left(\rho^{v}_{r,m} \cdot \mathrm{ONI}_{y,m}\right) = s_c\right]}_{\text{(ii) predicted direction}} \cdot \underbrace{\mathbf{1}\left[\,\mathrm{sign}\left(Z^{v,c}_{r,y,m}\right) = s_c\right]}_{\text{(iii) realised Z-anomaly}}. \tag{2}$$

where $s_c = +1$ for flood and heat and $s_c = -1$ for drought. The first condition requires an active NOAA ONI phase; the second requires the regional teleconnection to predict the channel-relevant direction; and the third requires the realized anomaly to move in that direction.

The ENSO-attributable climate shock flag is then AND-gate:

$$S^c_{i,y,m} = \mathbf{1}\big(\text{Threshold}^c_{i,y,m} = 1\big) \cdot \text{ENSOConsistent}^c_{r,y,m}. \tag{3}$$

where

$$\text{Threshold}^c_{i,y,m} = 1 \tag{4}$$

is the indicator that the operational threshold is crossed by a general climate shock. This flag and ENSO-consistency components are built independently and meet only in this AND gate.

Acute mine-month specification. Flood and heat are estimated at mine-month frequency:

$$\log(Output_{kilotonnes})_{i,y,m} = \alpha_i + \gamma_{r,m} + \delta_y + \beta X^c_{i,y,m} + \theta \log(\text{price})_{y,m} + \varepsilon_{i,y,m}. \tag{5}$$

Here α_i absorbs each mine's average production scale, $\gamma_{r,m}$ absorbs region-by-calendar-month seasonality, and δ_y absorbs common annual shocks. The monthly LME copper price controls for within-year price variation. The focal regressor $X^c_{i,y,m}$ takes two forms, run separately:

- **General channel-specific shock** uses the operational threshold flag from eq. (4). It fires whenever the channel's operational threshold is crossed in mine-month (i,m), regardless of cause. $\hat{\beta}^A_c$ is the average effect of a threshold-firing month on log mine output.
- **ENSO-attributable channel-specific shock** uses the AND-gate flag from eq. (3). It fires only on the subset of threshold-firings that are also ENSO-attributed by the rule in eq. (2). $\hat{\beta}^B_c$ is the effect of the ENSO-attributed subset against a comparison group that pools non-ENSO threshold months with genuinely quiet months.

Slow mine-year drought specification. Drought requires a different approach from flood and heat because both the measure and the climate mechanism are persistent. For this slow channel, the gate is interpreted as a persistent climate-state filter rather than a purely contemporaneous one-month attribution rule: the drought month must cross the SPEI-12 threshold and align with the regional ENSO teleconnection in the drought direction. SPEI-12 is a twelve-month standardized meteorological water-balance index: a monthly drought observation already summarizes the previous year of precipitation deficits and evaporative demand. The monthly drought flags are first converted into an annual mine-level exposure share: the fraction of valid months in year y in which mine i is in SPEI-12 drought. The output regression then uses the lagged three-year exposure history, rather than the contemporaneous drought share, because water scarcity affects production with delay and because current-year operating decisions and water-management responses may move together. The mine-year equation is

$$\log(Output_{kilotonnes})_{i,y} = \alpha_i + \delta_y + \beta X^{\text{DR}}_{i,y} + \varepsilon_{i,y} \tag{6}$$

where $X^{\text{DR}}_{i,y}$ is the three-year cumulative exposure share over years $y-1$ to $y-3$.

Inference and robustness. The Chilean mining panel has only seven regional clusters, so all acute estimates are interpreted through a small-cluster lens. Alongside region-clustered standard errors, we use restricted, null-imposed wild-cluster bootstrap-t inference, leave-one-region-out refits, leave-one-event-out refits for the major ENSO episodes, and a sensitivity battery that replaces the baseline rolling ONI and Pearson teleconnection with seasonal-window ONI, Spearman ρ, and phase-conditional anomaly rules. For drought, where the exposure is concentrated in the late megadrought, the evidence is read as a package: fixed-effects ladder, matched-window teleconnections, gate-definition sensitivity, lead placebo, leave-one-region-out, and randomization inference.

4.2 Desal-Adaptation Model: Desalination and Output

The moderated specification. We now ask whether commissioning a desalination plant moderates a mine's output response to ENSO-related stress. We let the stress effect vary with a mine's adaptation status by augmenting the impact specification of section 4.1 with an interaction term:

$$\log(Output_{kilotonnes})_{i,t} = \alpha_i + \gamma_{r,m(t)} + \delta_{y(t)} + \beta_1 \text{Stress}^c_{i,r,t} + \beta_2 \left(\text{Stress}^c_{i,r,t} \times \text{Adapt}_{i,t}\right) + \beta_3 \text{Adapt}_{i,t} + \theta \log(p_t) + \varepsilon_{i,r,t}. \tag{7}$$

Here β_1 is the stress effect on a mine that has not adapted; β_2 is the moderation—how the stress effect changes once a mine desalinates, so that cushioning implies $\beta_2 > 0$ and the effect on an adapted mine is $\beta_1 + \beta_2$; and β_3 absorbs the level difference in output between adapted and unadapted mines in normal months. Retaining β_3 is essential: omitting it forces the adapted–unadapted level gap to load onto β_2 and biases the moderation estimate.

The adaptation variable. We set $\text{Adapt}_{i,t} = 1$ from the first month a mine's dominant desalinated process is operational. As firm-level water consumption is confidential and not disclosed in the annual reports, desalination commissioning is the only adaptation signal we can observe; the flag is therefore a proxy for the water-source switch rather than a direct measure of it—a limitation we return to in the results. Commissioning dates are coded from operator annual reports (CMF *Memorias*), cross-checked against COCHILCO's public water-consumption studies and the ACADES desalination registry (section 3.2). Nine of the nineteen operations have commissioned some desalinated supply by the end of the sample, of which we code eight as adapted; four of the eight switch within the panel window (Spence, 2021; Quebrada Blanca and Collahuasi, 2023; Los Pelambres, 2024). The mines that rely on continental water throughout form the comparison group that identifies β_1. We report a binary headline coding alongside a switcher-only variant and a continuous share variant as sensitivity checks.

A built-in placebo. Desalination addresses water scarcity, so the water-source switch has a clear mechanism for the drought channel but none for flood or heat. For the *panel* interaction this gives a built-in check: if β_2 isolated the water switch, it would be positive for drought and near zero for the flood and heat placebos. A non-zero interaction in a no-mechanism channel would instead indicate that the panel β_2 is picking up mine *type*—which mines select into adaptation—rather than the water switch. This check concerns the panel

interaction; the within-mine event study we turn to in section 5.3 is a separate object, because it observes the *bundled* commissioning event—desalination together with the concentrator and process upgrades installed alongside it—rather than the water switch in isolation.

As the panel interaction has limited power for this question, we complement it with a within-mine event study around the commissioning dates; we set out that comparison, and the bundled-treatment caveat it carries, with the results in section 5.3.

Inference and robustness. As in the climate shocks model, the interaction is read through a small-cluster lens: with seven regional clusters we report restricted wild-cluster bootstrap-t p-values (Webb weights, 9,999 replications) alongside the asymptotic ones—the finer six-point scheme spans a far larger draw space than the two-point Rademacher weights, so we use more replications here than for the acute estimates—and we check the estimate against the three adaptation codings (binary, switcher-only, continuous share) and leave-one-mine-out and leave-one-region-out refits; the full battery is collected in section A.2.

5 Results

5.1 Event Accounting

The applied Climate Shocks Model methodology, described in section 4.1, allows us to account for climate-shock events as an annual share of months affected by general shocks, and an annual share of ENSO-attributed months. Figure 1 below shows the annual event accounting, with El Niño and La Niña episodes and the megadrought marked, and the 2006–2025 copper-mining output mapped above. The figure shows that the megadrought confounds almost the entire copper-mining output sample. There is no obvious co-movement between the climate shocks and output in the flood and heat channels, but the decline in output that began in 2018—tracking the rising share of unusually dry months, and further worsened by COVID-19—illustrates the lagged nature of the drought channel and its relevance to copper-mining output.

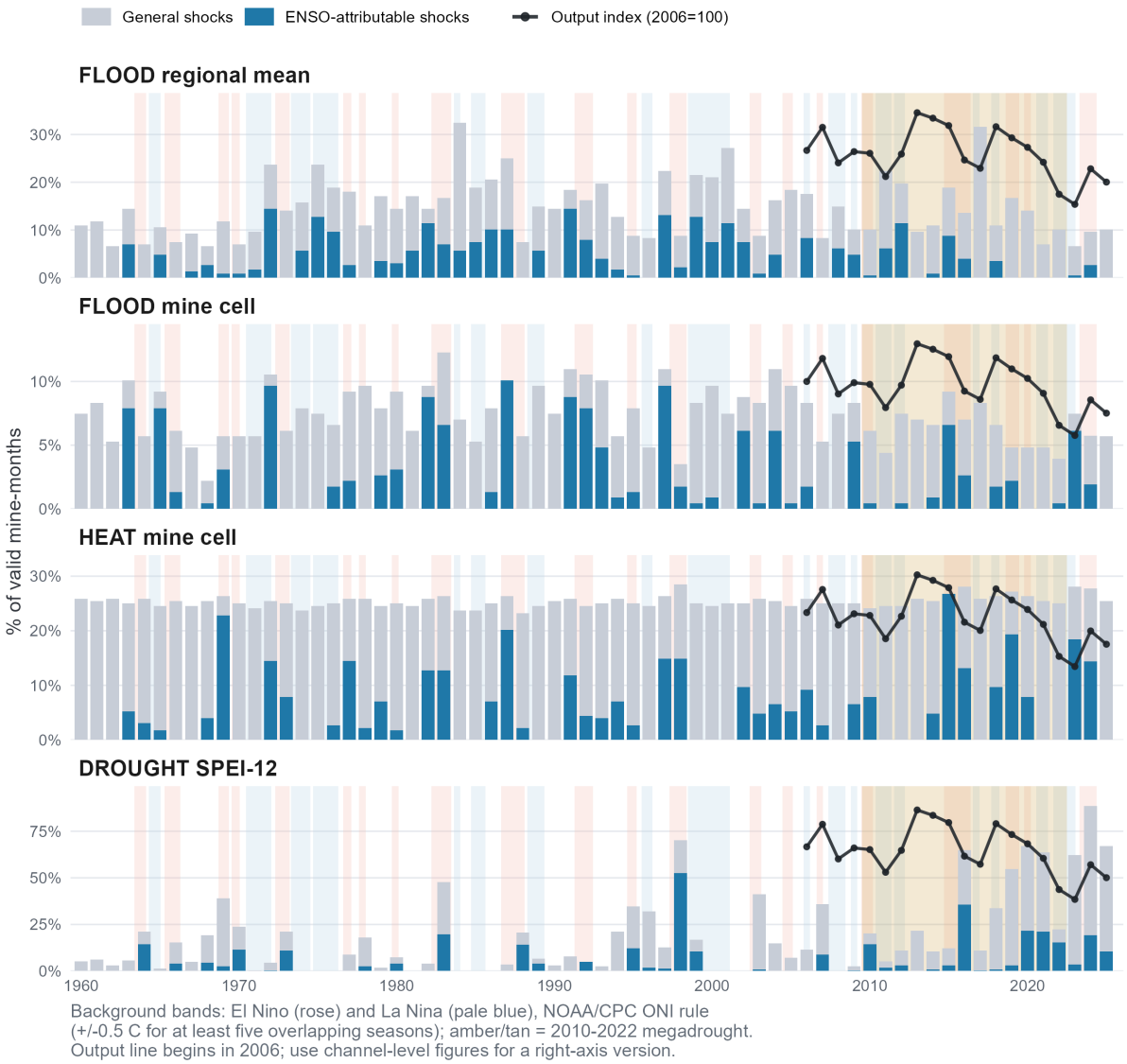


Figure 1. Annual event accounting across the main shock channels, with aggregate mine output overlaid. Blue bars are threshold-crossing observations that also pass the ENSO attribution rule; gray bars are threshold-crossing observations not classified as ENSO-attributable. Background bands show El Niño and La Niña episodes under the NOAA/CPC ONI rule; the amber band marks the 2010–2022 megadrought. The black line is the nineteen-mine annual copper-output index, normalized to 2006=100 and scaled within each panel for visual comparison.

5.2 Impact of Climate Shocks on Copper Mining Output

Table 2. Acute mine-month regressions: general shocks versus ENSO-attributable shocks

Specification	$\hat{\beta}$	SE	p_{CR0}	p_{WCB}	N
Flood, regional-mean monthly					
General flood shocks	-0.047	0.010	0.002***	0.013**	4,393
ENSO-attributed flood shocks	-0.025	0.020	0.246	0.287	4,393
Flood, mine-cell daily					
General flood shocks	-0.018	0.016	0.301	0.395	4,393
ENSO-attributed flood shocks	-0.002	0.028	0.939	0.978	4,393
Heat, mine-cell monthly					
General heat shocks	-0.003	0.009	0.713	0.686	4,393
ENSO-attributed heat shocks	-0.018	0.009	0.087*	0.700	4,393

Notes. General shock fires on any physical operational-threshold crossing. ENSO-attributable shock fires only when the AND gate also holds: active ONI, teleconnection-predicted direction, realized Z-anomaly direction, and threshold crossing. CR0 is asymptotic region-clustered inference on seven clusters; WCB is the restricted wild-cluster bootstrap-t with Rademacher weights, null imposed, $B = 999$. *Significance:* $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The climate-impact results are channel-specific. Floods leave the clearest monthly output signature; heat is a null at monthly mine-output frequency; and drought appears only as persistent, ENSO-attributed water stress accumulated over several years. This pattern is consistent with the mechanisms described in the methodology section: acute disruptions are visible in the month they occur, while water scarcity works through a slower stock.

Flood. The headline regional-mean flood specification shows a stable physical output effect. A wet regional month lowers monthly mine output by about 4.7% ($p_{\text{WCB}} = 0.013$), and the leave-one-region range remains entirely negative. The ENSO-attributable subset is smaller, -2.5%, and not statistically resolved under wild-cluster inference, but it has the same sign and is roughly half the general flood effect. The conservative mine-cell variant also points negative for general shocks (-1.8%) but is too sparse to identify the ENSO-attributed subset separately. The substantive reading is therefore not that every local rain event shuts a mine down; it is that broad wet regional months reduce output, with the ENSO-attributed subset accounting for a meaningful but less precisely estimated share.

Heat. Heat has no robust monthly output effect. The mine-cell threshold fires frequently at the hot Antofagasta-basin operations, but those mines appear to operate around routine heat exposure: the general shock coefficient is essentially zero and the apparent signal for ENSO-attributed shock disappears under the wild-cluster bootstrap. The result should be read narrowly. Monthly copper output cannot test short-run labor productivity, absenteeism, or shift-level heat effects; it only shows that routine heat exposure is not visible in monthly fine-copper tonnage.

Drought. Drought is different from flood and heat because of its lagged nature. The three-year cumulative share of all SPEI-12 drought months is not associated with lower annual output ($\hat{\beta} = +0.075$, $p = 0.510$). The output penalty is concentrated in persistent drought months that also pass the matched ENSO-consistency gate: active ONI, teleconnection-predicted dry direction, and realized regional precipitation deficit. In the preferred mine-year specification with mine and year fixed effects, the coefficient is $\hat{\beta} = -0.277$ ($p = 0.025$), implying that a 10 percentage-point increase in the three-year ENSO-consistent drought-exposure share is associated with roughly 2.8% lower annual mine output.

Table 3. Drought: general shock versus ENSO-attributed shock

Specification	$\hat{\beta}$	SE	p_{CR0}	p_{WCB}	N
General drought shocks, SPEI-12	+0.075	0.108	0.510	0.752	369
ENSO-attributed drought shocks, SPEI-12	-0.277	0.094	0.025**	0.013**	369

Notes. General drought shocks are all SPEI-12 drought months accumulated into the three-year exposure share. ENSO-attributed drought is the SPEI-12 subset classified by the matched 12-month gate: active ONI, teleconnection-predicted dry direction with matched-window ρ, and realized regional precipitation deficit. CR0 is asymptotic region-clustered inference on seven clusters; WCB is the restricted wild-cluster bootstrap-t with Rademacher weights, null imposed, $B = 999$. *Significance:* $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The drought result is real but bounded. It is negative under matched-window teleconnections, survives gate-definition checks and a lead placebo, and remains negative in every leave-one-region refit. At the same time, randomization inference, reported in section A.1, gives a more conservative one-sided placebo p-value of 0.134. Therefore the result should be read as evidence of ENSO-attributed persistent megadrought stress in this sample, rather than evidence of clean ENSO-attributed shock impact.

5.3 Testing the Adaptation Margin

Estimating the moderated specification in eq. (7) on the full panel does not, on its own, identify whether desalination cushions output, and the pattern of estimates shows why. Table 4 reports the interaction by channel. In the acute channels β_2 is small and statistically indistinguishable from zero; for the headline region-mean flood channel it is, if anything, negative—in a channel where desalination has no mechanism. The drought channel is sensitive to how exposure is measured: on a same-year drought share the ENSO interaction is negative ($\hat{\beta}_2 = -0.36$, $p = 0.01$), but on the three-year cumulative measure the climate shocks model uses (section 4.1) it is a clean null ($\hat{\beta}_2 = +0.09$, $p = 0.90$), so the same-year negative reflects a temporal mismatch rather than a stable adaptation effect.

Table 4. Moderated interaction by channel: estimates and the channel-stratified placebo

Channel	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	Expectation for β_2
Drought, meteo (cum. 3-yr)	—	$+0.086$	—	mechanism ($\beta_2 > 0$)
Flood, region-mean	-0.015	-0.248	$+0.368$	placebo ($\beta_2 \approx 0$)
Flood, mine-cell	$+0.021$	-0.140	$+0.364$	placebo ($\beta_2 \approx 0$)
Heat, mine-cell	-0.017	$+0.015$	$+0.362$	placebo ($\beta_2 \approx 0$)

Notes. ENSO-attributable shocks, headline adaptation coding, Anglo American Norte excluded; seven regional clusters. The acute channels are the monthly panel of eq. (7); drought is the annual mine-year panel, so its β_1 and β_3 are not comparable and are omitted. None of the β_2 estimates is significant: the asymptotic p for the region-mean flood placebo is 0.09, rising to 0.23 under the restricted wild-cluster bootstrap (Webb weights, null imposed, 9,999 replications), and the drought β_2 has $p = 0.90$; Rademacher weights give materially identical values. The level term β_3 is large and positive across channels, showing that adapted mines differ in normal-month output—a gap the interaction must hold separate.

Three diagnostics. Three features of the setting explain why the panel interaction cannot settle the question. First, the treated cells are small: only thirteen stressed-and-adapted mine-months across four mines identify the region-mean flood interaction, and dropping a single mine moves its p-value from 0.003 to 0.50. Second, adaptation status is correlated with location: the adapted mines sit in the driest northern regions and so face more meteorological drought than the comparison mines (46% versus 29% of months), so the interaction partly reflects where a mine is rather than whether it desalinated. Third, the within-sample switchers commissioned desalination together with concentrator ramp-ups during the peak-megadrought years, so the switcher-only drought interaction ($\hat{\beta}_2 = -2.31$) mixes the water switch with capacity expansion.

The channel-stratified placebo. The placebo built into eq. (7) is consistent with this reading. The channel where desalination should matter—drought—is a clean null, while the only nominally marginal coefficient is the region-mean flood placebo, a channel where the water switch cannot operate; and that coefficient is no longer even marginal under the wild-cluster bootstrap appropriate for seven clusters. The interaction therefore tracks the composition of adapted mines rather than the water switch, as one would expect when the adaptation flag proxies mine type rather than measured water use. We accordingly do not interpret the panel β_2 as desalination moderating the output response.

A within-mine comparison points in the same direction but cannot establish that adaptation is the cause: at Spence—the only switcher with a long enough post-commissioning window—heat stress cut output by about 18% before its 2021 SGO commissioning and has little effect after. As the heat channel has no water mechanism, the Spence comparison bears on the bundled commissioning rather than on the water switch: the commissioning pairs desalination with a new concentrator and process control, so any heat improvement would reflect those process changes rather than water (section 4.2). We read it as suggestive support from a single mine, not as identification. The substantive conclusion is that output is the wrong place to look for this adaptation: it surfaces not in tonnes but on operating costs, capital expenditure, closure provisions, and the water-rights constraints firms face—the firm-level, institutional, and financial record we turn to in section 6.

6 Adaptation in Practice

6.1 Anticipatory responses to ENSO forecasts

Among the climate hazards facing Chilean copper mining, the ENSO cycle is distinctive in being partially forecastable months ahead, which invites a natural question: if firms can anticipate an ENSO event, do they prepare for it? A forecast carries value only insofar as it can alter a decision still open to the firm, and insofar as that decision changes the firm's exposure; accuracy and lead time are necessary but irrelevant if no meaningful options remain available. This section tests whether ENSO's predictability yields such value, looking for an anticipatory response on the three margins the data render observable—monthly output, the environmental-permitting record, and the State's water-scarcity decrees—and finds none. We interpret this not as evidence that the forecast is uninformative, but as a specified case in which a forecastable climate signal nonetheless carries limited decision value for both firms and government, for reasons the section develops.

The forecast signal and the impact season. Each month the U.S. Climate Prediction Center and the International Research Institute for Climate and Society (CPC/IRI) issue a publicly available consensus probabilistic forecast assigning, for every upcoming three-month season, a probability assigned to each ENSO phase. We assess the forecast not against the *onset* of an episode—whose precise timing is hard to forecast through the spring predictability barrier[1] but against the *impact season* that disrupts a mine's operations: the austral-winter rainy season (June–August), when flooding affects the cordillera mines, and the austral-summer dry season (December–February), when water becomes scarce for the northern mines. The impact season is the decision-relevant object, since a mine acts on the forecast for the season in which its operations are at risk, not on the onset date. Read against the impact season, a majority-probability signal was present for every one of the five episodes (Figure 2): the two El Niño flood seasons crossed 50% two to three months ahead (peaking above 90% before the episode), and the three La Niña drought seasons crossed 50% three to six months ahead.

Detecting a forecast response in copper output. We posit that a mine which anticipates an ENSO-induced loss might pull production forward and build stockpile before the season, and that response would appear as output rising into the forecast window. To measure this we express each mine's monthly copper output as a deviation from its own typical level in that calendar month, benchmarked on neutral (non-ENSO) years, which removes the regular seasonal profile. For each episode e we compare the mean deviation during the forecast window with the mean deviation in the months immediately preceding it, within the same year, and term the difference the *output ramp*:

$$R_e \;=\; \overline{\mathrm{dev}}_{\,\mathrm{window}}(e) \;-\; \overline{\mathrm{dev}}_{\,\mathrm{before}}(e). \tag{8}$$

[1]The spring predictability barrier is the well-documented, abrupt drop in ENSO forecast skill for forecasts initialized in or spanning the boreal spring (roughly February–May), observed across both statistical and dynamical models regardless of initialization month (Webster and Yang 1992; Zheng and Zhu 2010). It is commonly attributed to the weak ocean–atmosphere coupling and small, noise-dominated sea surface temperature anomalies of the equatorial Pacific at that time of year. The barrier degrades the deterministic call on an episode's onset.

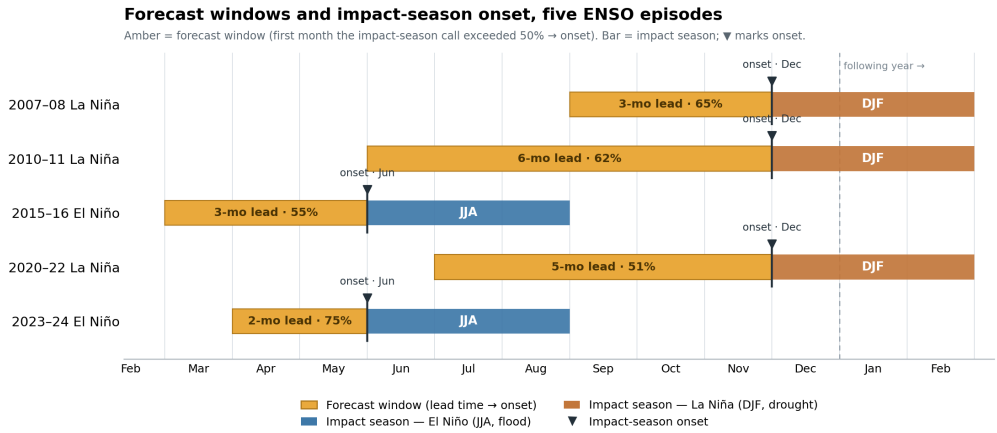


Figure 2. Forecast lead time to the impact season, five ENSO episodes. The amber window spans the first month the impact-season call exceeded 50% to its onset (▼); the bar is the impact season (blue JJA flood, brown DJF drought). The signal cleared 50% two to six months ahead in every episode.

We take each episode's forecast window and benchmark it against the calendar months preceding that 50% crossing, so the baseline is always drawn from the pre-signal period. In neutral years the ramp averages essentially zero; its year-to-year standard deviation defines the band (shown per episode in Figure B.3), and we count an episode as a forecast response only if it clears that band.

No episode shows anticipatory output ramping (Figure B.3): every episode lies within its band. The two El Niño flood episodes even disagree in sign (+1.3% in 2015 and −2.9% in 2023) and the three La Niña drought episodes scatter similarly (+0.9%, −2.5%, +3.9%), none approaching its band. The 2015 case is worth deeper consideration since it carried the earliest and clearest forecast in the sample. Its flood-mine output did run above normal through the window—but it ran above normal *all year*, by approximately +18% in January, two months before the forecast crossed 50% and five months before the episode realization (rainy season). 2015 was a high-output year for these mines, oscillating around a higher level rather than an output ramping building into the forecast window.

No response in the permitting or regulatory record. We extend this anticipatory investigation with two records of discrete firms and government decisions. If ENSO forecasts carry operational value to firms, one possible firm response is to accelerate water-related permitting when a forecast signals elevated flood or drought risk. We examine first, the e-SEIA environmental-filing record during each forecast window. Across all five episodes, not a single mine with exposure to the forecasted hazard—flood-prone cordillera operations during El Niño episodes or water-constrained northern operations during La Niña episodes—filed a water-supply or drought-adaptation permit. Desalination filings instead cluster after the 2022 Water Code reform. Second, the State's own *decretos de zona de escasez*, where the state regulator induces government powers to manage drought, would plausibly cluster ahead of the dry season. In reality the decrees confirm scarcity once it has matured rather than anticipating it, even when the ENSO signal is available months prior. Taking a multi-year view, the decrees track the deficit rather than any single season's ENSO state: activity

climbs through the 2014–2021 megadrought and then saturates, the El Niño years sitting among the quietest of all (Figure B.4).

Why the forecast goes unexploited: a slow tool and a weak signal. We tested flood and drought climate channels for an anticipatory response; on each, the forecast goes unused, but the source differs by channel. For drought the constraint is temporal. The principal defense against drought shocks is structural: mines secure supply by constructing desalination plants and coastal pipelines. The share of seawater in total mining water use increased from 11.6% in 2014 to 34.1% in 2024 (Figure B.5), and COCHILCO projects this share to approach 68% by 2034. Although deliberate, this transition answers to the region's long-run aridification and the 2022 Water Code (Section 6.2) far more than to any single forecast; and because infrastructure of this nature takes years to plan, permit, finance, and build, no signal that holds for mere weeks or months can reorient it. This reasoning extends to permitting and capital decisions, which turn on multi-year cycles.

For flooding, the binding constraint seems to lie in the signal itself. A confident impact-season ENSO call remains a weak predictor of whether a flood will in fact impact at a given, often high-altitude, mine. On neither channel, then, does the predictability that sets ENSO apart from most climate hazards confer much practical advantage: for drought the adaptation tool is too slow, for flood the mine-level signal too weak—even though the forecast is, in principle, available and accurate enough to act upon. It bears emphasizing that this analysis observes only output, public filings, and the decree registry; the finer, short-term adjustments—rescheduling maintenance, pre-positioning equipment, contracting for water—lie beyond our data, and cannot on that account be ruled out. Our cautious claim is that ENSO-relatedness offers no observable anticipatory advantage on the channels we are able to observe, although our analysis is clearly constrained by our small sample size and data access.

6.2 Institutional and Water-Governance Response

History of Chile's Water Code. Prior to 1981, water in Chile was managed primarily through state-administered concessions. The 1969 Water Code strengthened government control by treating water as a public resource and limiting private transfers of water rights (Ríos and Quiroz 1995). The 1981 Water Code fundamentally changed this approach by separating water rights from land ownership, establishing private property rights over water, and allowing rights to be freely traded. The reform aimed to improve allocative efficiency, encourage investment, and reduce state involvement in water management by relying on market mechanisms to direct water toward its highest-value use (Bauer 1997; Prieto et al. 2019). While proponents argued that water markets improved flexibility and economic efficiency, critics contended that high transaction costs, weak oversight, and unequal access limited market performance and failed to address environmental and social externalities (Bauer 1997; Boelens 2009).

A further challenge emerged through the over-allocation of water rights, often referred to as "paper water". Under the 1981 framework, the Dirección General de Aguas (DGA), Chile's national water authority responsible for administering and allocating water rights had limited authority to deny applications when rights appeared legally available, resulting in the issuance of water entitlements that exceeded physically available resources in several basins and aquifers. Consequently, the volume of legally granted rights often

surpassed actual hydrological availability, creating structural imbalances between paper entitlements and real water supplies (Prieto et al. 2019).

Push Towards Reform. Three pressures converged to force institutional change. First, the 2010–2022 megadrought reduced water availability across the mining north and agricultural centre. Second, the state was unable to reallocate effectively: constitutional protection of private rights left the DGA able to act only under formally declared scarcity, while basins were already legally exhausted. Third, the desiccation of the Loa wetlands and the dispossession of Atacameño and peasant communities, alongside rural drinking-water crises, exposed water commodification's social costs (Prieto et al. 2019; Suárez Delucchi 2024).

In response, the DGA sought to recover authority. The 2005 amendment introduced the *patente por no uso* — an escalating fee on idle rights — and required applicants to justify intended use (Donoso 2015). The decisive shift came with the 2022 Water Code Reform; it ended perpetual ownership, converting rights into 30-year revocable concessions, and established a statutory priority cascade placing human consumption and ecosystem flow above productive use, with new powers to prohibit rights over glaciers, create reserves, and expropriate for domestic supply (Macpherson et al. 2023). The DGA's existing instruments — aquifer restriction and prohibition zones, river exhaustion declarations, and scarcity decrees — became the operational map of where rights now exceed supply (Appendix B.7, panel 1).

Impact on the mining sector. The mining sector came under focus from these evolving governance arrangements. Although Chile did not enact a law explicitly requiring mines to use desalinated water, legislative efforts sought to accelerate the sector's transition away from continental water sources [2]. In addition, increasing restrictions on freshwater extraction, stricter environmental permitting requirements by the Resolución de Calificación Ambiental (RCA), enhanced DGA oversight, and growing social opposition to groundwater use collectively encouraged mining companies to invest in desalination infrastructure. The experience of the Copiapó and Loa basins, where water rights overallocation and groundwater depletion generated significant conflicts among mining companies, agricultural users, and local communities, contributed to a broader policy shift toward desalination as a strategic source of water supply for northern Chilean mining operations (Prieto et al. 2019; Rivera et al. 2016). The spatial overlap shows that 13 of 19 major copper mines operate inside or within 10 km of a legally restricted or prohibited aquifer or an exhausted river basin (Appendix B.6). The firm response to move to desalination is a rational one to ensure sustainability of future activities. Reflecting this transition, the share of seawater in total mining water use rose from 11.6% in 2014 to 34.1% in 2024, with the rise concentrated in the northern mining regions (Appendix B.8).

Cerro Colorado (BHP, Tarapacá) is the warning where over-extraction from the Lagunillas aquifer damaged protected wetlands and the mine eventually closed in 2023. The institutional environment no longer merely permits adaptation; it imposes the cost of inaction on firms still dependent on continental freshwater.

[2] Boletín 10.038-08 proposed amendments to the Mining Code requiring greater use of seawater in mining operations while obligating the state to promote and facilitate seawater and desalination infrastructure

6.3 Footprint of Water Stress: Costs, Provisions, and the Balance Sheet

Section 5 finds a muted response to ENSO-related drought stress. We now attempt to uncover where the effect lands, using operator financial statements over 2006–2024. This follows the same approach from Pankratz et al. (2023) and Addoum et al. (2023) — output changes little, so the effect should appear in income statement and balance sheet. The evidence is descriptive and selective over 14 operators covering 19 mines (Table B.10) and with uneven cost disclosures, annual and financial reports.

We extract general flood and drought, ENSO-attributed flood and drought shocks [3] defined in Section 4, and we extend the analysis to key financial measures in the annual reports. A snapshot of the recorded shock events are reflected in Figure B.10 and Figure 3 below.

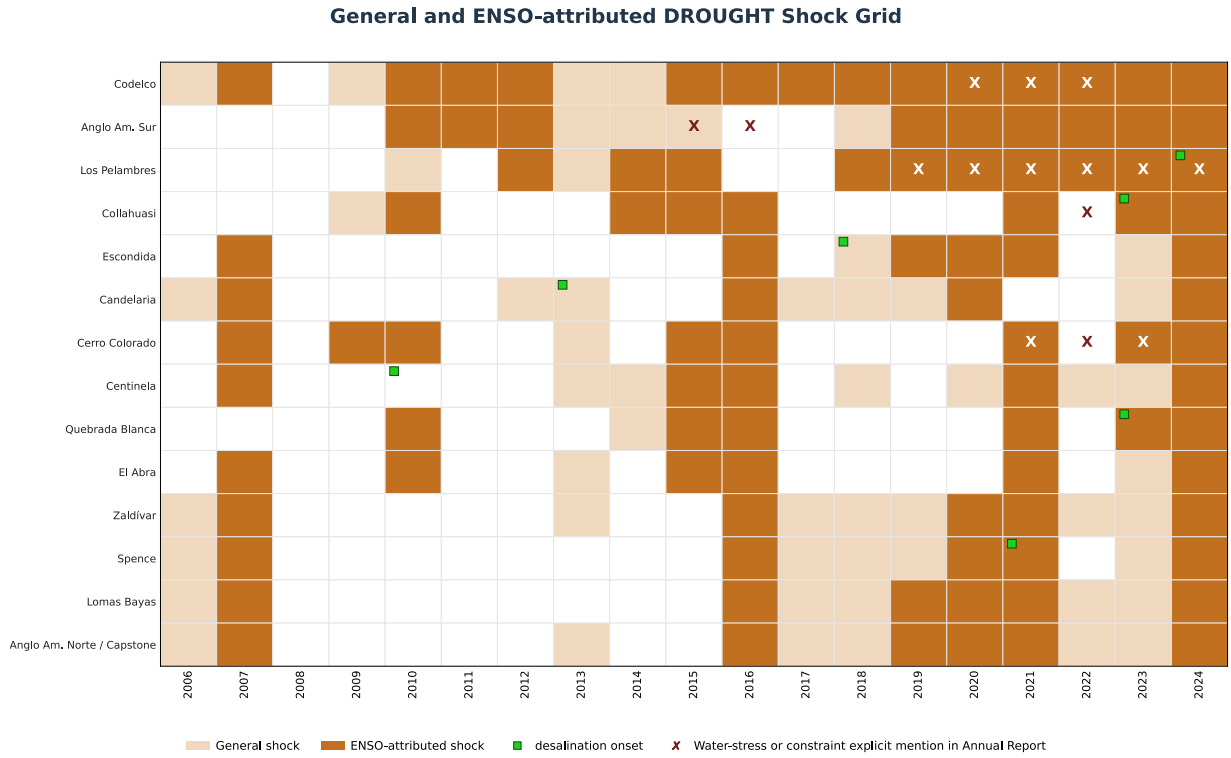


Figure 3. Drought shock grid, 2006–2024. Light cells recorded a general drought shock (SPEI-12 ≤ -1) only; dark cells recorded an ENSO-attributed drought shock. An **X** marks a company-year with a water-stress mention in the annual report; row labels in teal denote desalinated operations (filled square marks onset).

Flood. Of all mine-months that cross the flood threshold, these floods are brief: when a mine records one in a given year it lasts a single month in 60% of cases (Figure B.10)—too short to register in a full fiscal year. The closest to a visible case is Anglo American Sur, whose C1[4] rose 33% at El Soldado (173 to 230 ¢/lb)

[3] For the two operators that aggregate several mines in the climate panel—Codelco (five divisions) and Centinela (El Tesoro and Esperanza)—a company-year is flagged if *any* constituent mine crosses the drought threshold in *any* month (a union across mines and months). As the financial statements are consolidated, this union matches the reporting unit. Flood-cell counts are distinct firing-months.

[4] C1 (cash) cost is the standardised industry measure of the direct cost of producing one pound of copper—mining, processing, on-site administration, and freight and refining charges. It is reported in US cents per pound (US¢/lb).

and 23% at Los Bronces (67 to 83 ¢/lb) in the 2009 flood shock, while group output held steady. These observations are not sufficient to ascertain a clean impact on the financial statements.

Drought — Production & Operating Cost Impact. We identify the impact of drought shocks using two widespread drought episodes, in 2016 and 2021, that affected nearly all copper operators (12 and 13 of the 14 operators, respectively). These episodes occurred at opposite ends of the copper price cycle—when prices were relatively low in 2016 and near historic highs in 2021 (Figure B.9)—helping to separate drought-related operating cost effects from commodity price movements. In 2016, amid sector-wide cost pressures, output falls in several mines (El Abra −32%, Escondida −13%, Quebrada Blanca −11%, Candelaria −10%). In 2021, despite record prices, several mines still saw a fall in production (Escondida −16%, Los Pelambres −10%, Zaldívar −9%) and pay more per pound (Zaldívar C1 +33%, Los Pelambres +10%). Management corroborates: Anglo American attributes Los Bronces' lower output to "reduced water availability, following the record low levels of precipitation in 2021 and 2022," with C1 cash costs up 35% (to 214 ¢/lb) driven partly by "higher water management costs"[5].

Long Term Debt and Capital Expenditure. Once management commits to a desalination project, the response to water scarcity becomes a structural balance-sheet adjustment rather than an operating-flow response. The key financial signals are CAPEX[6] and non-current liabilities. Across desalinating operations, a familiar pattern emerges: CAPEX rises ahead of commissioning and long-term debt increases to fund it.

Quebrada Blanca is the clearest case—its 100%-seawater QB2 project lifted non-current liabilities from US$3.5bn (2020) to US$12.0bn (2024). Centinela, built on seawater from the outset, took long-term debt from near zero to US$1.9bn in three years, and in 2024 secured a further US$2.5bn of 12-year project financing for its desalination-integrated *Nueva Centinela* expansion[7]; and Escondida's non-current liabilities rose US$2.4bn between 2015 and 2017, ahead of its 2018 commissioning—consistent with the operator's disclosure that its 2,500 L/s seawater-desalination plant required an investment of approximately US$3.4bn.[8] These headline costs often bundle the water infrastructure within broader concentrator and mine-expansion programmes rather than isolating the desalination plant. The pattern suggests that adaptation converts a potentially recurring and volatile operating shock into a one-time, debt-financed capital stock. In doing so, exposure shifts from the operating statement to the balance sheet, leaving firms more leveraged to become less vulnerable to future water shortages.

Annual-report disclosures. Operators' own water-stress disclosures cover 15 mine-years across five operations. Los Pelambres records a severe Coquimbo drought, reduced Choapa-river flows and its *Plan*

[5]At Los Bronces, "C1 unit costs increased by 35% to 214 c/lb (2021: 158 c/lb), driven by high inflation, planned lower production and higher water management costs," reflecting "reduced water availability, following the record low levels of precipitation in 2021 and 2022." Anglo American plc, *Integrated Annual Report 2022*, p. 92.

[6]Capital expenditure is spending to acquire, build, or upgrade long-lived physical assets—here, capitalised on the balance sheet rather than expensed.

[7]Minera Centinela, *Memoria Anual* 2024, Hechos Relevantes, p. 9: "Se firma el acuerdo de financiamiento del proyecto Nueva Centinela por US$2.500 millones. . . El préstamo fue pactado a 12 años, con un período de gracia de cuatro años."

[8]Minera Escondida, *Memoria Anual* 2017, sección de proyectos: "demandó una inversión aproximada de US$3.400 millones"; the plant supplements an earlier 525 L/s unit commissioned in 2006.

Sequía across 2019–2024, with C1 increasing year-on-year (+10% in 2021, +24% in 2022)[9]. Collahuasi, citing Tarapacá water restrictions and the start of its desalination build, shows the sharpest single jump (+43% in 2022).[10]

Escondida — the desalination benchmark. Escondida, the world's largest copper-producing operation, commissioned its first seawater desalination plant at Puerto Coloso in 2006 and subsequently invested US$3.4bn[8] in a second facility, operational by 2017, to eliminate aquifer dependency. Despite carrying seven ENSO-attributed drought years across the study period, the operation produced no water-stress disclosure in any annual report and recorded positive net income in every year of the panel (2008–2024, ranging from US$993m in 2016 to US$4.3bn in 2010). Its throughput did fall in shock years — 13% in 2016 and 15% in 2021 — but as a desalinated operation these movements track ore grade and mine sequencing rather than water availability, and none of the balance-sheet distress markers appear: no going-concern qualification, no erosion of equity, no provision-driven conversion of asset into liability. The absence of a financial stress reflects a capital commitment made two decades ahead of the shocks it was designed to absorb.

Cerro Colorado — provisions, closure and exit. For operations lacking that backstop, sustained water scarcity is reflected directly in balance-sheet provisions. Cerro Colorado — a BHP-operated copper-cathode producer in the Tarapacá region dependent on the continental Lagunillas aquifer — is the most fully documented instance. Following a sequence of rulings by the First Environmental Tribunal of Antofagasta from September 2021, initially suspending and subsequently restricting aquifer extraction, the financial statements deteriorate in step: a closure provision[11] that rose 3.1× from US$266m in 2019 to US$838m in 2023, total equity falling from US$163m to negative US$803m over the same period, and CAPEX collapsing from US$167m to US$9.5m. The going-concern notes in the 2022 and 2023 statements cite the expiry of the mine's environmental licence at end-2023 — the regulatory terminus of the aquifer-restriction process — as the basis for uncertainty, and production ceased in late 2023. The water channel here is best characterised as contributory rather than singular: low ore grades and weak prices were independently pushing the asset toward the end of its economic life. Drought stress did not initiate the decline so much as foreclose any extension of it, accelerating the conversion of a productive asset into an environmental liability.

7 Limitations

Several limitations constrain the study, spanning what is measureable, what the design can identify, and how far the findings generalize. The most crucial is a data constraint: COCHILCO's firm-level water-consumption

[9] Antofagasta plc, *Annual Report 2022*, water and operating-review sections, and Minera Los Pelambres financial statements filed with the CMF; Antofagasta reports reduced water availability in the Choapa valley through the 2019–2022 drought and the supply measures culminating in the INCO desalination plant (commissioned 2024).

[10] Compañía Minera Doña Inés de Collahuasi SCM, financial statements filed with the CMF; the Tarapacá water-restriction context and the Patache desalination programme (phased from 2023) are set out in Collahuasi's sustainability reporting and in the annual reports of its owners Anglo American and Glencore.

[11] A closure provision is the estimated future cost of dismantling a mine and rehabilitating its site, recognised under IAS 37 and re-measured as expected costs and remaining mine life evolve. At Cerro Colorado, it encompasses aquifer-restoration obligations, making it a direct financial representation of accumulated water-related environmental liability.

microdata (EMPAE) are confidential, so we cannot observe each mine's seawater share month to month, which would be the most precise element to empirically identify adaptation. The commissioning dates are the closest observable proxy—though the public COCHILCO aggregates corroborate the broad pattern, with the sector's seawater use rising from a low single-digit share of mining water in the mid-2000s to roughly a third by the early 2020s, concentrated in the arid north. Inference is a second constraint: the acute estimates rest on only seven regional clusters, so all of it runs through a small-cluster lens that the wild-cluster bootstrap mitigates but cannot remove. Monthly output is also a coarse margin for the slow and routine channels, the regional-mean flood effect is robust, but the drought effect is carried largely by the 2010s megadrought (randomization inference $p = 0.134$) and reads as megadrought stress rather than a general drought response, while the heat null shows only that routine heat does not surface in monthly tonnage, not that it is costless. Relatedly, the adaptation channel is not identified on output: the desalination interaction is underpowered—it flips on dropping a single mine—and confounded by selection, since the adapted mines sit in the driest north, while the adaptation flag proxies commissioning rather than the confidential seawater share and the within-sample switchers bundled desalination with new concentrators; we therefore document the absorption of climate stress on costs, capital, and provisions rather than estimating it on output. The anticipatory test is bounded in turn: it rests on five ENSO episodes—too few for distributional inference and sensitive to any single event—and observes only output, public filings, and the decree registry, not short-run adjustments such as procurement, maintenance scheduling, and water contracting, so its null is that no anticipatory response is *observable*, not that none occurs. Finally, external validity is narrow: nineteen operations in one country and one commodity, with an adapted group of large, well-capitalised incumbents that could finance desalination early—a selected set whose experience need not generalise to smaller operators or other mining jurisdictions.

8 Conclusion

Our findings reveal a complex, cautious answer to our question, "Does Chilean copper mining sector react differently to climate shocks induced by ENSO?". First, while the share of ENSO-attributed months is visible in each channel, it is significant and statistically resolved only for the drought channel: an 10% increased share of the ENSO-related drought months in a prior 3-year window leads to a 2.8% loss of output.
The strongest result for flood channel (-4.7% of output for each unusually wet month on regional level) is not ENSO-attributable. Second, the recurring nature of ENSO implies the specific forecasting opportunities, but there is no public evidence that government and firms are using it for anticipation. The reaction on acute shocks (flood and heat) is situational, and there is clear time mismatch between forecasting of drought and available institutional and operational mechanisms of long-term response.
Finally, the lack of strong evidence of output response to climate shocks combined with the clear picture of government measures for water management and footprints of adaptation to water-scarcity in the firms financial statements is telling us, that adaptation of Chilean copper mining sector to constant climate challenges is an ongoing story started long ago - and ENSO is only a part of it.

References

Jawad M. Addoum, David T. Ng, and Ariel Ortiz-Bobea. Temperature shocks and industry earnings news. *Journal of Financial Economics*, 150(1):1–45, 2023. doi: 10.1016/j.jfineco.2023.07.002.

Bebonchu Atems and Naafey Sardar. Exploring asymmetries in the effects of ENSO on U.S. food and agricultural stock prices. *The Quarterly Review of Economics and Finance*, 2021.

Alan Barreca, Karen Clay, Olivier Deschênes, Michael Greenstone, and Joseph S. Shapiro. Adapting to climate change: The remarkable decline in the U.S. temperature–mortality relationship over the twentieth century. *Journal of Political Economy*, 124(1):105–159, 2016.

Carl J. Bauer. Bringing water markets down to earth: The political economy of water rights in Chile, 1976–95. *World Development*, 25(5):639–656, 1997.

Felipe Beltrán, Luigi Durand, Mario González-Frugone, and Javier Moreno. A preliminary assessment of the economic effects of climate change in chile. Working Paper 997, Central Bank of Chile, 2023.

Rutgerd Boelens. The politics of disciplining water rights. *Development and Change*, 40(2):307–331, 2009.

Marshall Burke and Kyle Emerick. Adaptation to climate change: Evidence from US agriculture. *American Economic Journal: Economic Policy*, 8(3):106–140, 2016.

Serhan Cevik and João Tovar Jalles. Eye of the storm: The impact of climate shocks on inflation and growth. IMF Working Paper WP/23/87, International Monetary Fund, 2023.

Comisión Chilena del Cobre (COCHILCO). Anuario de estadísticas del cobre y otros minerales. Technical report, COCHILCO, Santiago, Chile, 2024.

Olivier Damette, Clément Mathonnat, and Julien Thavard. Climate and sovereign risk: The latin american experience with strong ENSO events. *World Development*, 178:106590, 2024.

Olivier Damette, Maxime Fajeau, Rémi Generoso, and Clément Mathonnat. Climate shocks and U.S. bank stability. Working Paper hal-05396142, HAL, 2025.

J. Ignacio Del Rio, Paulina Fernandez, Emilio Castillo, and Luis Felipe Orellana. Assessing climate change risk in the mining industry: A case study in the copper industry in the antofagasta region, chile. *Commodities*, 2(3):1–15, 2023.

Melissa Dell, Benjamin F. Jones, and Benjamin A. Olken. What do we learn from the weather? the new climate–economy literature. *Journal of Economic Literature*, 52(3):740–798, 2014.

Guillermo Donoso. Chilean water rights markets as a water allocation mechanism. In Manuel Lago, Jaroslav Mysiak, Carlos M. Gómez, Gonzalo Delacámara, and Alexandros Maziotis, editors, *Use of Economic Instruments in Water Policy: Insights from International Experience*, volume 14 of *Global Issues in Water Policy*, chapter 19. Springer, Cham, Switzerland, 2015.

Gilles Dufrénot, William Ginn, and Marc Pourroy. Climate pattern effects on global economic conditions. *Economic Modelling*, 141, 2024.

René D. Garreaud, Camila Alvarez-Garreton, Jonathan Barichivich, Juan Pablo Boisier, Duncan Christie, Mauricio Galleguillos, Carlos LeQuesne, James McPhee, and Mauricio Zambrano-Bigiarini. The 2010–2015 megadrought in central Chile: impacts on regional hydroclimate and vegetation. *Hydrology and Earth System Sciences*, 21(12):6307–6327, 2017. doi: 10.5194/hess-21-6307-2017.

Karla Hernández and Carlos Madeira. The impact of climate change on economic output in chile: Past and future. Documentos de trabajo, Central Bank of Chile, 2021.

Benjamin F. Jones and Benjamin A. Olken. Climate shocks and exports. *American Economic Review: Papers and Proceedings*, 100(2):454–459, 2010.

Charles D. Kolstad and Frances C. Moore. Estimating the economic impacts of climate change using weather observations. NBER Working Paper 25537, National Bureau of Economic Research, 2018.

Yi Liu, Wenju Cai, Xiaopei Lin, Ziguang Li, and Ying Zhang. Nonlinear el niño impacts on the global economy under climate change. *Nature Communications*, 14(1):1–11, 2023.

Xueting Luo, Ke Gong, Aixing Li, Xiaomei Ding, and Yuhang Yang. The impact of ENSO shocks on firm performance: The role of supply chain resilience and network complexity in energy firms. *Working paper*, 2026.

Elizabeth Macpherson, Cristy Clark, Pía Weber Salazar, Natalie Baird, Afshin Akhtar-Khavari, and Edward Challies. Evolving rights to (and of) water in Chile: A case for relationship-based water law and governance. *The International Journal of Human Rights*, 27:1–26, 2023. doi: 10.1080/13642987.2023.2266719.

Robert Mendelsohn, William D. Nordhaus, and Daigee Shaw. The impact of global warming on agriculture: A ricardian analysis. *American Economic Review*, 84(4):753–771, 1994.

Clément Nedoncelle and Julien Wolfersberger. Weather shocks and firm exports in developing countries. *Review of World Economics*, 161:743–784, 2025.

NOAA Climate Prediction Center. Cold and warm episodes by season (oceanic Niño index). https://origin.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ONI_v5.php, 2024.

Scott D. Odell, Anthony Bebbington, and Karen E. Frey. Mining and climate change: A review and framework for analysis. *The Extractive Industries and Society*, 5(1):201–214, 2018. doi: 10.1016/j.exis.2017.12.004.

Nora Pankratz, Rob Bauer, and Jeroen Derwall. Climate change, firm performance, and investor surprises. *Management Science*, 69(12):7352–7398, 2023. doi: 10.1287/mnsc.2023.4685.

Manuel Prieto, María Christina Fragkou, and Matías Calderón. Water policy and management in Chile. In Patricia A. Maurice, editor, *Encyclopedia of Water: Science, Technology, and Society*. Wiley, 2019.

Mónica A. Ríos and Jorge A. Quiroz. The market of water rights in Chile: Major issues. *Cuadernos de Economía*, 32(97):317–345, 1995.

Diego Rivera, Alex Godoy-Faúndez, Mario Lillo, Amaya Alvez, Verónica Delgado, Consuelo Gonzalo-Martín, Ernestina Menasalvas, Roberto Costumero, and Ángel García-Pedrero. Legal disputes as a proxy for regional conflicts over water rights in Chile. *Journal of Hydrology*, 535:36–45, 2016.

Adriana Angela Suárez Delucchi. Unpacking 'community water management' in rural Chile: An institutional ethnography. *Journal of Rural Studies*, 110:103345, 2024.

Yongping Sun, Ying Yang, Nan Huang, and Xin Zou. The impacts of climate change risks on financial performance of mining industry: Evidence from listed companies in china. *Resources Policy*, 69, 2020.

David Ubilava. The ENSO effect and asymmetries in wheat price dynamics. *World Development*, 96:490–502, 2017. doi: 10.1016/j.worlddev.2017.03.031.

U.S. Geological Survey. Mineral commodity summaries 2024: Copper. Technical report, U.S. Geological Survey, Reston, VA, 2024.

Sergio M. Vicente-Serrano, Santiago Beguería, and Juan I. López-Moreno. A multiscalar drought index sensitive to global warming: The Standardized Precipitation Evapotranspiration Index. *Journal of Climate*, 23(7):1696–1718, 2010. doi: 10.1175/2009JCLI2909.1.

Andrew C. Wilcox, Cristían Escauriaza, Roberto Agredano, Emmanuel Mignot, Vicente Zuazo, Sebastián Otárola, Luca Castro, Jorge Gironás, Rodrigo Cienfuegos, and Luca Mao. An integrated analysis of the March 2015 Atacama floods. *Geophysical Research Letters*, 43(15):8035–8043, 2016. doi: 10.1002/2016GL069751.

A Robustness Checks

A.1 Climate Shocks Model

Small-cluster inference and influence checks. The acute regressions have only seven regional clusters, so the headline interpretation does not rest on asymptotic clustered p-values alone. The regional-mean flood effect is the only acute coefficient that remains clearly resolved under the wild-cluster bootstrap, and it is also stable under leave-one-region and leave-one-event refits. The narrower ENSO-attributable flood coefficient is sign-stable but not statistically resolved once the firing count is reduced by the AND gate.

Table A.1. Influence checks for the acute output-impact estimates

Specification	$\hat{\beta}$	p_{WCB}	Leave-one-region range	Leave-one-event range
Flood reg-mean, general shock	-0.047	0.013	[-0.058, -0.042]	[-0.051, -0.040]
Flood reg-mean, ENSO-attributable shock	-0.025	0.287	[-0.038, -0.013]	[-0.053, -0.018]
Flood mine-cell, general shock	-0.018	0.395	[-0.023, -0.007]	[-0.018, -0.012]
Flood mine-cell, ENSO-attributable shock	-0.002	0.978	[-0.018, 0.024]	[-0.023, 0.031]
Heat mine-cell, general shock	-0.003	0.686	[-0.031, -0.000]	[-0.015, 0.019]
Heat mine-cell, ENSO-attributable shock	-0.018	0.700	[-0.099, -0.014]	[-0.048, -0.001]

Notes. Leave-one-event refits drop the 2015–2016 El Niño, 2020–2022 La Niña, and 2023–2024 El Niño windows one at a time. The heat ENSO-attributable-shock coefficient is sign-stable in these refits but fails the wild-cluster bootstrap, which is why it is treated as a null in the main text.

ENSO-attribution rule sensitivity. For flood and heat, the baseline AND gate uses rolling 3-month ONI, a Pearson regional teleconnection, and the realised Z anomaly. For drought, the baseline row is the final matched-window Pearson-ρ specification used in the headline drought table, so that the robustness annex and main text refer to the same estimator. Three alternatives test whether the results depend mechanically on the attribution rule: a seasonal-window ONI (JJA for flood and drought, DJF for heat), a Spearman teleconnection, and a phase-conditional mean-anomaly direction. The flood ENSO-attributable-shock coefficient remains negative under all alternatives; heat remains a null; and drought remains negative except under the seasonal-window variant, which is less appropriate for a slow accumulator because it discards within-year variation.

Table A.2. ENSO-attribution robustness

Specification	Attribution rule	$\hat{\beta}$	SE	p_{CR0}
Flood reg-mean ENSO-attributable shock	Baseline: 3-mo ONI, Pearson ρ	-0.025	0.020	0.246
Flood reg-mean ENSO-attributable shock	Seasonal-window ONI	-0.062	0.025	0.048
Flood reg-mean ENSO-attributable shock	Spearman ρ	-0.038	0.017	0.065
Flood reg-mean ENSO-attributable shock	Phase-conditional $\overline{Z}$	-0.034	0.021	0.159
Heat mine-cell ENSO-attributable shock	Seasonal-window ONI	-0.015	0.016	0.378
Heat mine-cell ENSO-attributable shock	Spearman ρ	-0.017	0.011	0.162
Heat mine-cell ENSO-attributable shock	Phase-conditional $\overline{Z}$	-0.016	0.011	0.210
Drought SPEI-12	Matched-window Pearson ρ (headline)	-0.277	0.094	0.025
Drought SPEI-12	Seasonal-window ONI	-0.174	0.211	0.441
Drought SPEI-12	Spearman ρ	-0.246	0.060	0.006
Drought SPEI-12	Phase-conditional $\overline{Z}$	-0.278	0.053	0.002

Drought robustness and identification. The drought result is deliberately presented as an evidentiary package rather than as a single asymptotic p-value. The negative coefficient appears only when SPEI drought is classified by the matched ENSO-consistency gate; all-drought exposure is null. The coefficient is negative at multi-month SPEI accumulation scales, survives gate-definition checks, passes a lead placebo, and remains negative in all seven leave-one-region refits. The binding caveat is that the effect is carried by the post-2020 late-megadrought block.

Table A.3. Drought inference and influence checks

Specification	$\hat{\beta}$	SE	p_{CR0}	p_{WCB}	RI p	LOO-region range	N
All-drought SPEI-12	+0.075	0.108	0.510	0.752	–	[-0.050, +0.182]	369
ENSO-consistent matched-ρ SPEI-12	-0.277	0.094	0.025**	0.013**	0.134	[-0.443, -0.211]	369

Notes. LOO-region range reports the minimum and maximum coefficient across the seven leave-one-region refits. The WCB column uses a restricted wild-cluster bootstrap-t with Rademacher weights, null imposed, and $B = 999$. RI is a one-sided randomisation-inference placebo for the ENSO-consistent drought design ($B = 2000$); the corresponding two-sided value is 0.263. RI is not reported for the all-drought row because that row is the general drought exposure, not the event-concentrated ENSO attribution design.
*Significance: *p < 0.10, **p < 0.05, ***p < 0.01.*

Table A.4. Drought robustness battery

Specification	$\hat{\beta}$	p_{CR0}	Reading
All-drought SPEI-12	+0.075	0.510	Threshold-only drought is null.
ENSO-consistent matched-ρ SPEI-12	-0.277	0.025	Preferred estimate.
SPEI-3 with matched 3-month gate	-0.429	0.037	Negative at multi-month scale.
SPEI-6 with matched 6-month gate	-0.447	0.061	Negative, marginal.
Drop realised-precipitation condition	-0.266	0.024	Barely moves.
Drop teleconnection direction	-0.101	0.212	Teleconnection condition is essential.
Realised dryness only	+0.071	0.501	No ENSO gate, null.
Pre-2020 output years only	+0.801	0.131	No pre-megadrought negative signal.
Post-2020 output years only	-0.615	0.051	Identification block.
Lead placebo, future exposure	+0.094	0.549	Future drought does not predict current output.

Notes. Robustness rows use the same mine-year output panel unless otherwise stated. The lead placebo replaces current three-year drought exposure with future exposure. Inference and influence diagnostics for the two headline drought specifications are reported in Table A.3.

Hydrological SSI-12 cross-check. SSI-12 was explored as a hydrological drought cross-index but is not retained in the main specification. It measures source-basin streamflow rather than mine-cell meteorological water balance, and representative gauges are unavailable or physically inappropriate for many operations: closed endorheic salars, desalinated or direct-seawater operations, and coastal/cross-region aggregates. On the small gauged-basin subset, the matched ENSO-consistent SSI-12 coefficient is positive rather than negative ($\hat{\beta} = +0.501$, $p = 0.065$, $N = 148$). This does not overturn the SPEI-12 result, but it does limit the claim: the drought finding rests on meteorological water balance, with streamflow treated as a non-corroborating triangulation check.

A.2 Desal-Adaptation Model

This appendix collects the robustness battery behind the adaptation results in section 5.3. The message is consistent across every check: the moderated interaction is underpowered and confounded, and no specification isolates a desalination effect on output.

Full acute-interaction estimates. Table A.5 reports the complete coefficient set summarized in the body, for the headline binary coding (Adapt = `main`). The interaction β_2 is negative in both flood channels and near zero for heat, and none is clearly distinguishable from zero. The level term β_3 is large, positive, and similar across channels, confirming that adapted mines differ in normal-month output—a difference that must be absorbed separately rather than through the interaction.

Table A.5. Acute moderated interaction, full coefficients

Channel	β_1 (p)	β_2 (p)	β_3 (p)	$\beta_1+\beta_2$
Flood, mine-cell	+0.021 (0.56)	−0.140 (0.49)	+0.364 (0.06)	−0.119
Flood, region-mean	−0.015 (0.54)	−0.248 (0.09)	+0.368 (0.06)	−0.263
Heat, mine-cell	−0.017 (0.47)	+0.015 (0.88)	+0.362 (0.06)	−0.002

Notes. ENSO-attributed shock, headline adaptation coding, Anglo American Norte excluded. Cluster-robust p-values at the region level (7 clusters) in parentheses.

Temporal specification of the drought interaction. The drought interaction turns entirely on the temporal definition of the regressor. Table A.6 shows that the same-year drought share—used in an earlier draft—returns a significant negative interaction, but once aligned to the three-year cumulative measure the Climate Shocks Model uses, the effect is a clean null. The switcher-only restriction returns a large negative coefficient, but it is identified off the four within-sample switchers, whose plants were commissioned bundled with concentrator ramp-ups during the peak megadrought, and so confounds the water switch with capacity expansion.

Table A.6. Drought moderation by temporal specification and sample

Regressor	Sample	β_2	p
Cumulative 3-yr, ENSO-aligned	Full	+0.09	0.90
Cumulative 3-yr, all-cause	Full	+0.05	0.71
Cumulative 3-yr, ENSO-aligned	Switcher-only	−2.31	0.03
Same-year share (superseded)	Full	−0.36	0.01

Notes. Annual mine-year panel, mine and year fixed effects. The switcher-only sample retains only the four within-sample switchers; its estimate is confounded by bundled concentrator commissioning (see below).

Leave-one-mine-out fragility. The single marginal coefficient—the region-mean flood placebo—is fragile to the sample. Table A.7 drops each early-adapted mine in turn: the estimate ranges from -0.08 ($p = 0.50$) to -0.38 ($p = 0.003$), so one mine moves it from insignificant to highly significant. A coefficient this sensitive to a single observation cannot support an adaptation reading, especially in a channel where desalination has no mechanism.

Table A.7. Leave-one-mine-out: region-mean flood interaction β_2

Sample	β_2	p
Full sample	−0.25	0.089
Drop Esperanza	−0.08	0.499
Drop Candelaria	−0.38	0.003
Drop El Tesoro	−0.36	0.183

Notes. Headline coding, ENSO-attributed flood region-mean channel. Each row re-estimates the interaction dropping one early-adapted mine.

Alternative adaptation codings. The conclusions are unchanged under the alternative codings. The strict (switcher-only) variant excludes the four early adopters and is the basis for the switcher-only row of table A.6; the continuous share variant, which assigns intermediate weight to the partially desalinated operations (Collahuasi, Los Pelambres), leaves both the drought null and the marginal, fragile flood placebo intact.

Selection and treatment-cell sizes. Two facts explain the pattern. Adapted mines face systematically more meteorological drought than unadapted mines (46% versus 29% of months) and similar hydrological drought, because they sit in the driest northern regions—which is why they desalinated; the interaction therefore reflects composition rather than moderation. And the treated cells are very small: thirteen stressed-and-adapted mine-months across four mines for region-mean flood, forty-six across three mines for heat, and only two for mine-cell flood.

Small-cluster inference. With only seven regional clusters we report wild-cluster-bootstrap p-values alongside the asymptotic ones in the channel-stratified placebo (table 4). The bootstrap (restricted, Webb weights, 9,999 replications) leaves the point estimates unchanged and moves the marginal flood placebo from $p = 0.09$ to $p = 0.23$; Rademacher weights give materially identical values. Under proper small-cluster inference no channel is significant.

The role of the level term β_3. Throughout we retain the adaptation level term β_3. Omitting it forces the normal-month output gap between adapted and unadapted mines—which is large and positive ($\beta_3 \approx +0.36$)—onto the interaction, mechanically biasing β_2. The estimates above are not robust to dropping β_3, nor should they be: doing so is a misspecification, not a robustness check.

Bundled treatment. Finally, two of the four within-sample switchers (Spence's SGO and Quebrada Blanca's QB2) commissioned desalination jointly with a new concentrator, so for those mines the post-commissioning period bundles a water-source change with capacity expansion and process modernization.

This is why we do not lean on the switcher-only drought coefficient, and why identification rests on the within-mine event study of section 5.3 rather than the panel interaction.

B Additional Results and Figures

B.1 Data

The mine-level desalination registry and adaptation coding summarized in section 3.2 are reported in full in table B.8.

Table B.8. Mine-level desalination registry and adaptation coding

Mine	Onset	Source	Class
Esperanza	2010	Centinela district (direct seawater)	early
El Tesoro	2010	Centinela district (direct seawater)	early
Candelaria	2013	Own plant (500 l/s)	early
Escondida	2018	EWS (own; 100% by 2020)	early
Spence	2021	SGO (own; 1,000 l/s)	switcher
Quebrada Blanca	2023	QB2 (own; 1,200 l/s)	switcher
Collahuasi	2023	Patache (shared, phased)	switcher
Los Pelambres	2024	INCO (own; 400 l/s)	switcher
Never adapted in sample (comparison group): Codelco Gaby, Salvador, Andina, Teniente, Chuquicamata/RT/MH; El Abra, Zaldívar, Lomas Bayas, Anglo American Sur, Cerro Colorado. *Excluded:* Anglo American Norte (no desalination in cell).			

Notes. Onset is the first year the mine's dominant process runs on desalinated or direct seawater. "Early" adopters entered the sample adapted or switched before the analysis window; "switcher" commissionings are observed within the panel and identify the event study.

B.2 Climate Anomalies and Event Accounting

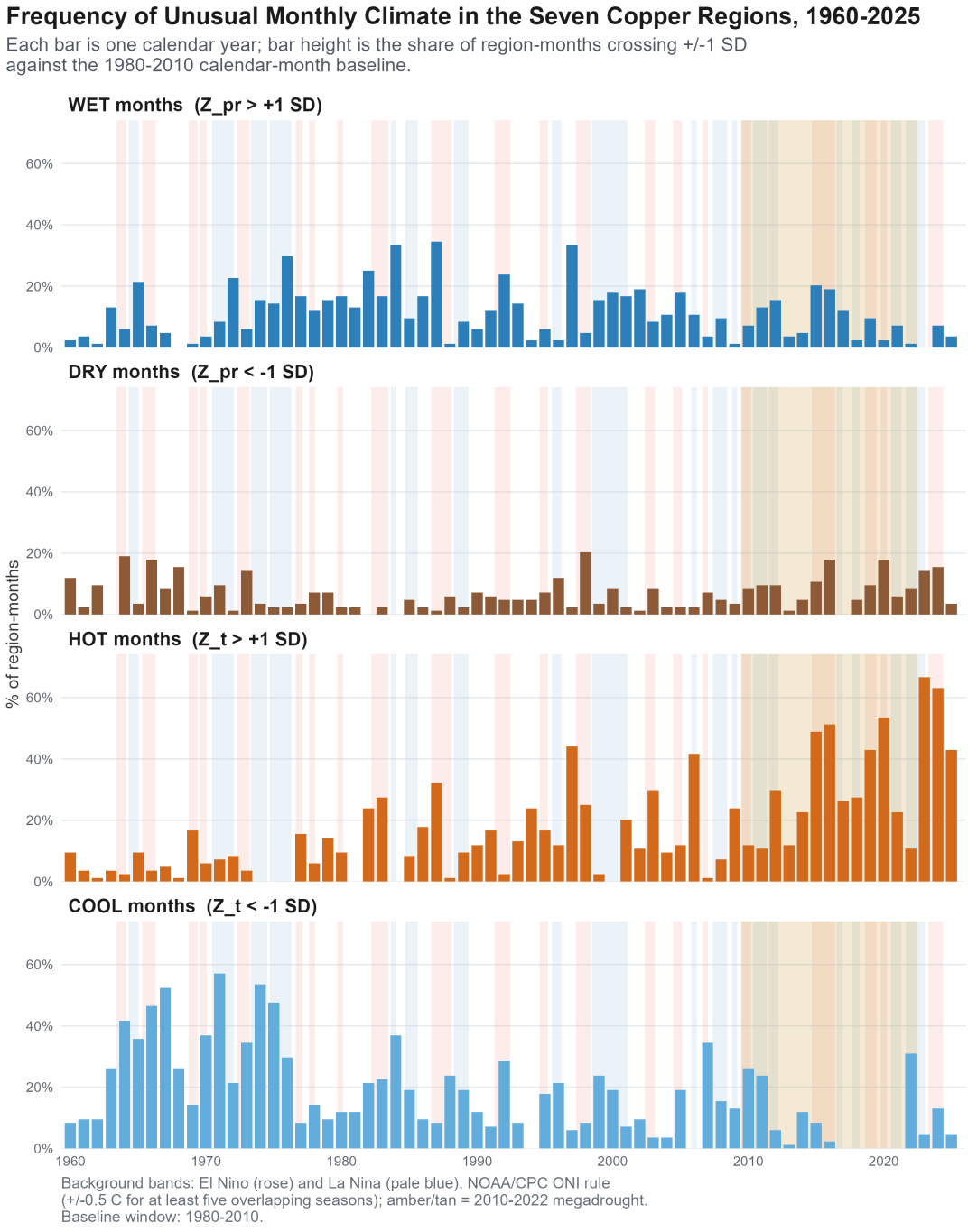


Figure B.1. Frequency of unusual monthly climate in the seven copper regions, 1960–2025. Bars show the annual share of region-months with precipitation or temperature anomalies exceeding one standard deviation relative to the 1980–2010 calendar-month baseline. Rose and pale-blue background bands show NOAA/CPC historical ONI El Niño and La Niña episodes, defined by the $\pm 0.5^{\circ}$C ONI threshold persisting for at least five overlapping seasons; amber/tan marks the 2010–2022 megadrought.

Stress-Flag Accounting The ENSO-attributable stress flag is intentionally narrower than the general shock flag. A general shock can occur without satisfying the ENSO-consistency rule, and an active ENSO teleconnection can occur without crossing an operational threshold. The stress flag is only the intersection of the two.

Table B.9. General and ENSO-attributable shock firings in the master stress ledger

Channel	Valid rows	General shocks	%	ENSO shocks	%
Flood, regional mean	15,048	2,206	14.66	639	4.25
Flood, mine cell	15,029	1,114	7.41	334	2.22
Heat, mine cell	15,029	3,816	25.39	838	5.58
Drought, SPEI-12	14,839	2,780	18.73	776	5.23

Notes. Counts come from the mine $\times$ month $\times$ channel stress ledger.

B.3 Output Impact of Climate Shocks

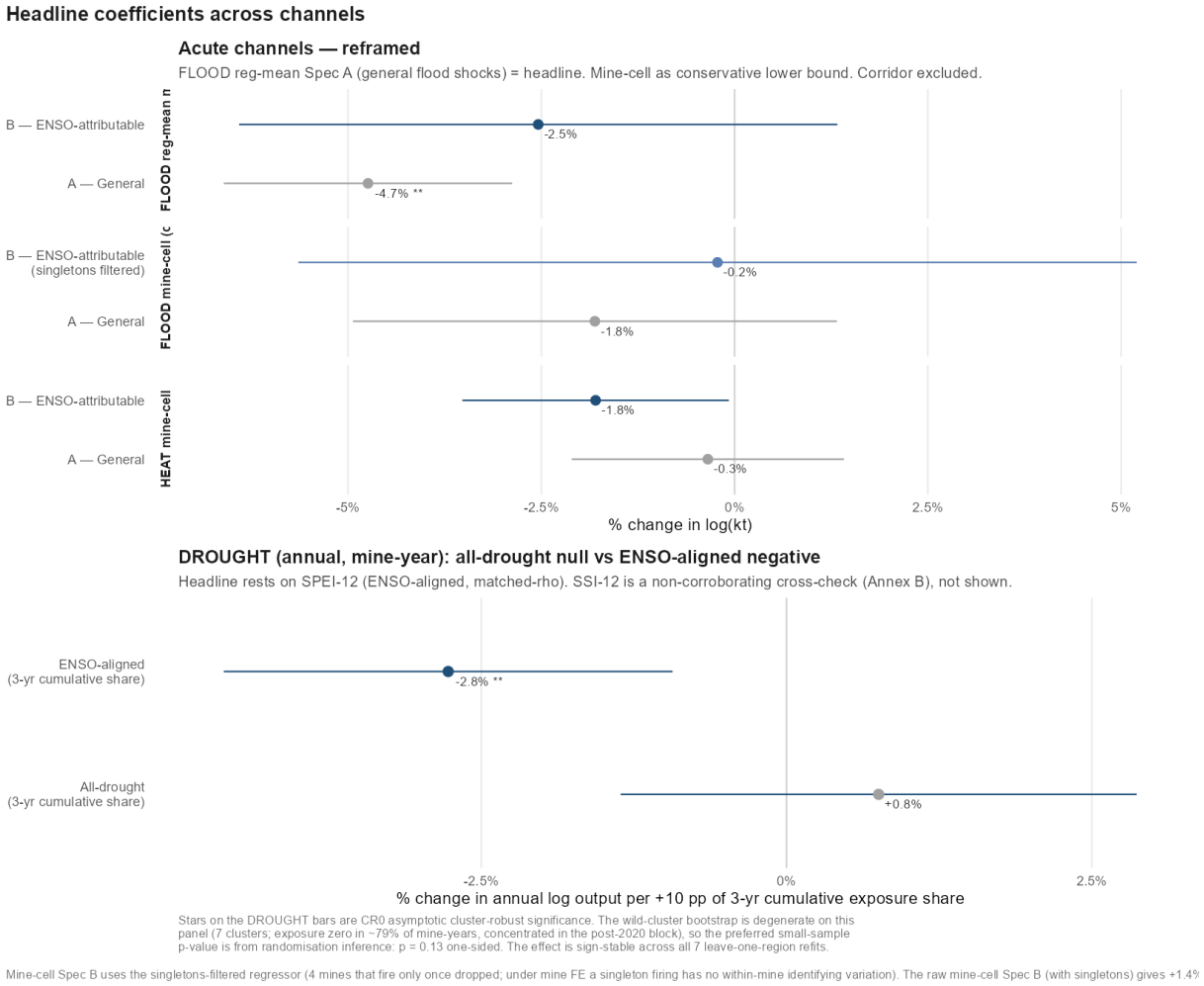


Figure B.2. Forest plot of headline coefficients. Acute rows are changes in monthly log mine output for a 0-to-1 stress-flag firing. Drought rows are changes in annual log output for a 10 percentage-point increase in the three-year cumulative drought-exposure share.

The forest plot shows why the results chapter treats the three channels separately. Flood is an acute, month-level disruption with a clearly negative general-shock coefficient; heat is a monthly output null; drought is a slow annual exposure whose coefficient is not directly comparable to the binary acute rows because it scales with cumulative exposure intensity.

B.4 Firm-Level Response and ENSO Forecasts

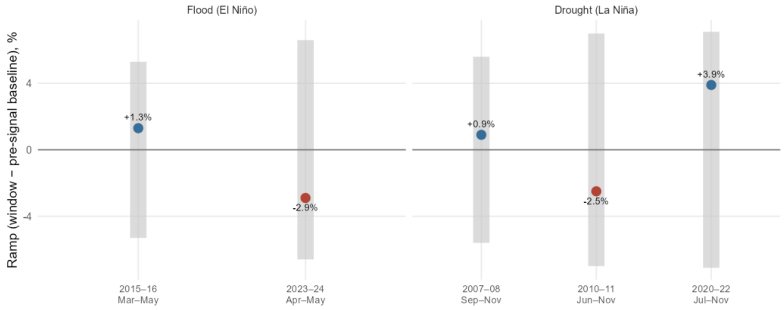


Figure B.3. Output ramp for each ENSO episode (point) against its own ±1 s.d. neutral-year band (gray bar). The ramp is the mean output deviation in the forecast lead window (shown beneath each label) minus the pre-signal baseline—the in-year months before the impact-season forecast crossed 50 percent—with deviations taken from the neutral-ONI-year mean for the same mine and calendar month, so mine level, ordinary seasonality, and a same-year level shift net out. Ramp and band are both in percent of log output. Every episode lies within its band. The two El Niño panels are the anticipation test proper; the three La Niña panels, where adaptation is infrastructural rather than an output decision, are shown for completeness.

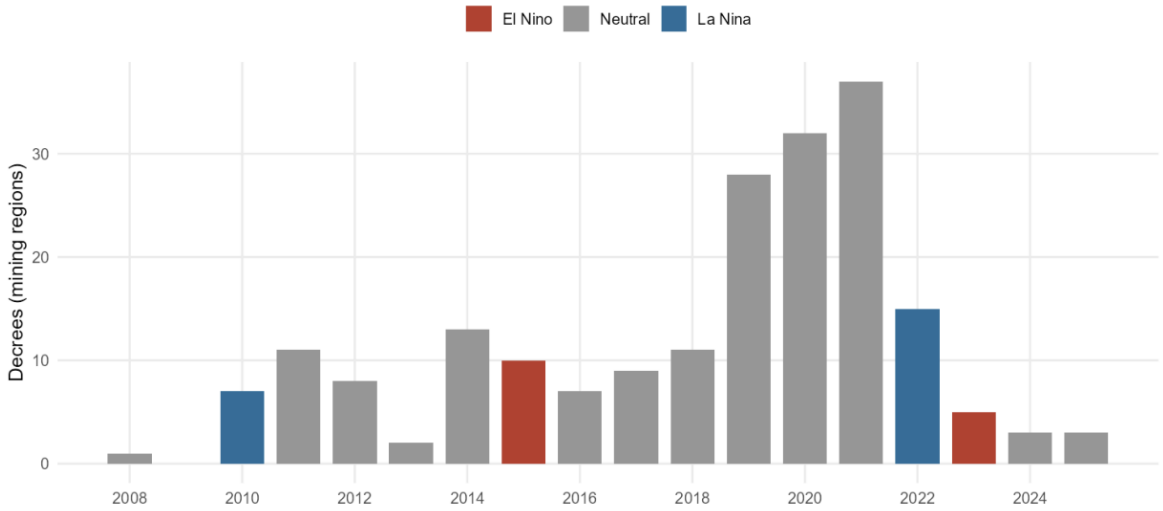


Figure B.4. DGA water-scarcity decrees in the seven mining regions, by water year (2008–2025), colored by the July-centered (JJA) ENSO phase of that year's recharge winter. Decree activity is flat through the early years, climbs steeply with the 2014–2021 megadrought, and saturates—tracking the multi-year structural deficit rather than any single year's ENSO state. The phase coloring shows no anticipatory pattern: the peak years are overwhelmingly ENSO-neutral, and the two El Niño years (2015, 2023) are among the quietest in the record. The regulator's response is reactive and lagging, not forecast-led. Source: DGA decree registry; NOAA CPC ONI.

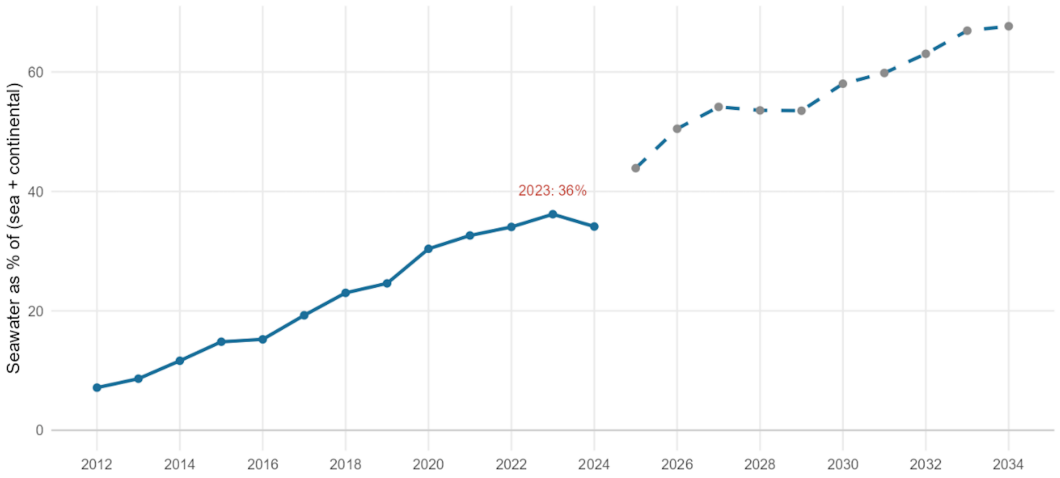


Figure B.5. Seawater share of copper-mining water supply. The solid line is the realised share, 2012–2024; the *dotted* continuation, 2025–2034, is the COCHILCO projection. The share climbs from about 7% in 2012 to 36% in 2023 and is projected to approach 68% by 2034—a structural, decade-scale substitution keyed to the region's long-run aridification and the 2022 Water Code rather than to any single ENSO forecast. *Source:* COCHILCO water-consumption panel.

B.5 Institutional and Water-Governance Response

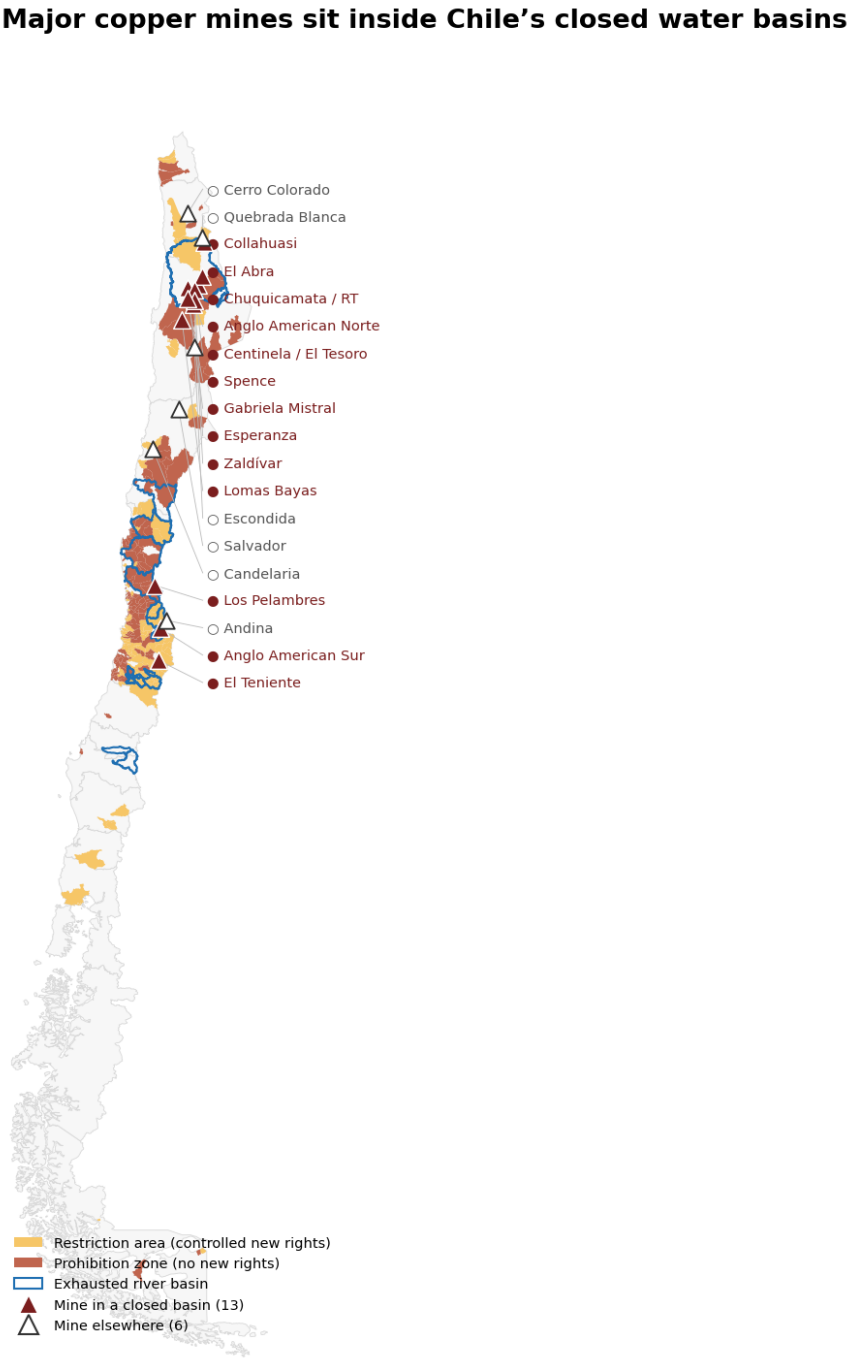


Figure B.6. Major copper mines inside Chile's closed water basins. Chile's nineteen largest copper operations are overlaid on legally constrained water bodies: aquifer restriction areas (controlled new rights), prohibition zones (no new rights), and exhausted river basins. Thirteen of the nineteen mines lie within such a closed basin, and sixteen within 10 km of one, showing that most major operations already face binding limits on new continental-water rights. Source: own elaboration; restriction, prohibition, and exhaustion layers from DGA Mapoteca Digital (November 2025); mine locations from the project registry ($n = 19$).

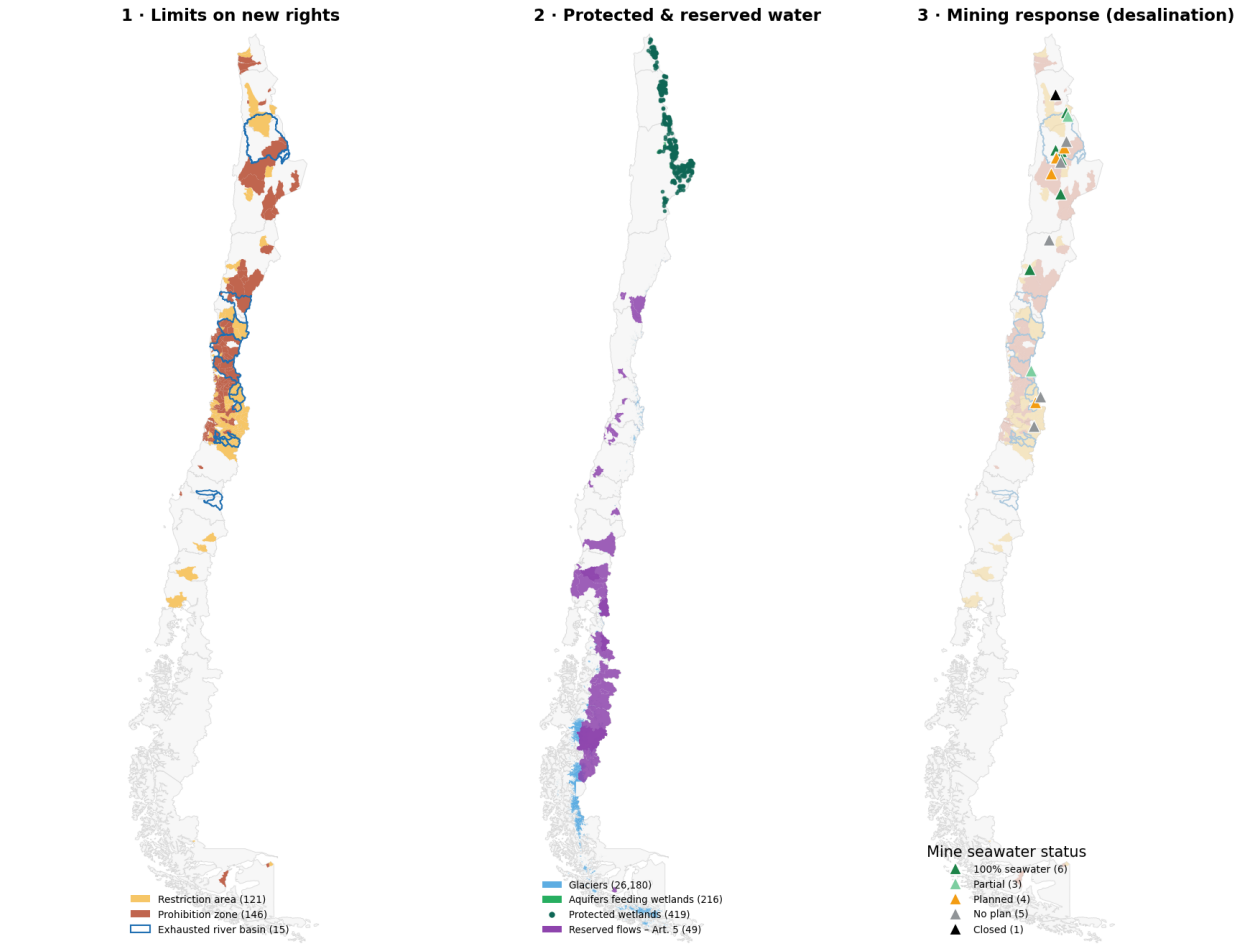


Figure B.7. Chile's water-governance constraints and the mining response. Panel 1 (*Limits on new rights*) maps aquifer restriction areas, prohibition zones, and exhausted river basins; Panel 2 (*Protected & reserved water*) maps glaciers, aquifers feeding protected wetlands, protected wetlands, and Article 5 reserved environmental flows; Panel 3 (*Mining response*) classifies each major copper mine by seawater status (fully on seawater, partial, planned, or none). Together the panels show that where continental water is legally constrained or protected, mines have moved—or are moving—to desalinated seawater. Source: governance layers from DGA Mapoteca Digital (November 2025); mine desalination status from the project registry ($n = 19$), shown at company-group level.

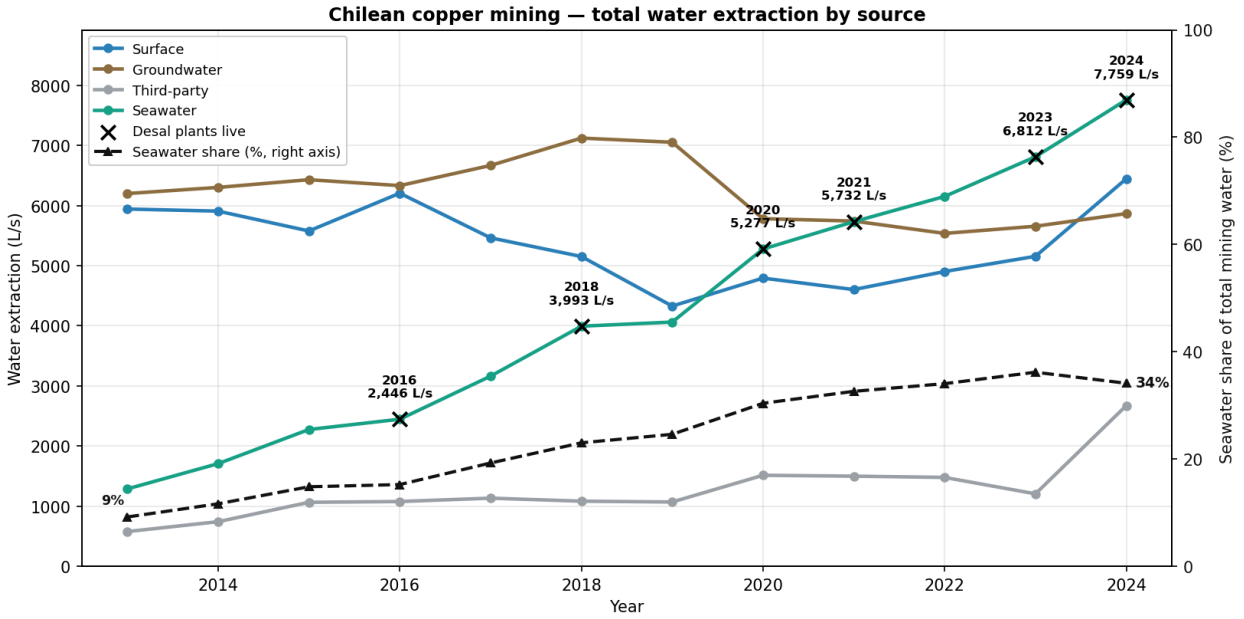


Figure B.8. Chilean copper mining: total water extraction by source, 2013–2024. The left axis shows withdrawals in L/s for each source (surface, groundwater, third-party, and seawater), with crosses (×) marking desalination-plant capacity coming on line. The right axis plots the seawater share of total mining water, which rises from 9% in 2013 to 34% in 2024 as continental freshwater (surface plus groundwater) falls from 87% to 54%. Source: COCHILCO Anuario, *Consumo de Agua en la Minería del Cobre* (2024).

B.6 Financial Evidence

Table B.10 maps the nineteen mine operations in the climate panel to the fourteen consolidated financial entities (CHC codes) used throughout. Codelco's five divisions are consolidated into a single financial entity (no divisional statements exist); El Tesoro and Esperanza are reported jointly as Centinela. The water-source column records whether each operation draws continental water or has transitioned to desalinated/direct seawater, with the commissioning year.

Table B.10. Mapping of mines to consolidated financial entities

Mine (panel name)	Operator	Ownership	Water source
Codelco - Andina	Codelco	Codelco (state-owned, 100%)	Continental
Codelco - Chuqui, RT y MH	Codelco	Codelco (state-owned, 100%)	Continental
Codelco - Gaby	Codelco	Codelco (state-owned, 100%)	Continental
Codelco - Salvador	Codelco	Codelco (state-owned, 100%)	Continental
Codelco - Teniente	Codelco	Codelco (state-owned, 100%)	Continental
Anglo American Sur[a]	Anglo American	Anglo American 50.1% / Codelco 29.5% / Mitsubishi 20.4%[b]	Continental
Escondida	BHP	BHP 57.5% / Rio Tinto 30% / JECO 12.5%	Desalinated 2018
Collahuasi	Independent JV[c]	Anglo American 44% / Glencore 44% / JCR B.V. 12%[d]	Continental[†]
Los Pelambres	Antofagasta Minerals	Antofagasta plc 60% / Japanese consortium 40%[e]	Desalinated 2024
Candelaria	Lundin Mining	Lundin Mining 81.8% / Sumitomo 18.2%	Desalinated 2013
El Tesoro	Antofagasta Minerals	Antofagasta plc 70% / Marubeni 30%	Seawater (direct) 2010
Esperanza	Antofagasta Minerals	Antofagasta plc 70% / Marubeni 30%	Seawater (direct) 2010
Quebrada Blanca	Teck	Teck 60% / Sumitomo 30% / Codelco 10%[f]	Desalinated 2023
Spence	BHP	BHP (100%)	Desalinated 2021
El Abra	Freeport-McMoRan	Freeport-McMoRan 51% / Codelco 49%	Continental
Zaldivar	Antofagasta Minerals	Antofagasta plc 50% / Barrick 50%	Continental
Lomas Bayas	Glencore	Glencore (100%)	Continental
Cerro Colorado	BHP	BHP (100%)	Continental (closed 2023)
Anglo American Norte[g]	Capstone Mining	Capstone 100% (MB) / Capstone 70%, Mitsubishi Materials 30% (MV)[h]	Continental

[a] Panel entity covers El Soldado and Los Bronces; consolidated under Anglo American Sur S.A. [b] Mitsubishi holds stake via Anglo American Sur Investment Ltd.
[c] No single parent consolidates 100%. [d] JCR B.V. is 91.9% Mitsui & Co. (2022); formerly co-owned with JX Nippon.
[e] Japanese consortium: JX Nippon 15.8% / Mitsubishi Materials 10.0% / Marubeni 9.2% / Mitsubishi Corp 5.0%.
[f] QB2 restructuring (2018). Codelco acquired ENAMI's 10% non-funding interest in September 2024 for US$520M.
[g] Panel entity covers Mantos Blancos and Mantoverde; operated by Anglo American Norte until Nov 2015, then Mantos Copper / Capstone post-acquisition. [h] Mitsubishi Materials acquired 30% of Mantoverde in February 2020 for US$236M.
[†] C20+ Patache desalination plant first delivered water May 2025 (post-sample); classified continental throughout.
Sources: Company annual reports; COCHILCO mine registry; Antofagasta plc, BHP, Anglo American, Teck, Glencore, Lundin Mining, Capstone Copper investor disclosures (2006–2024).

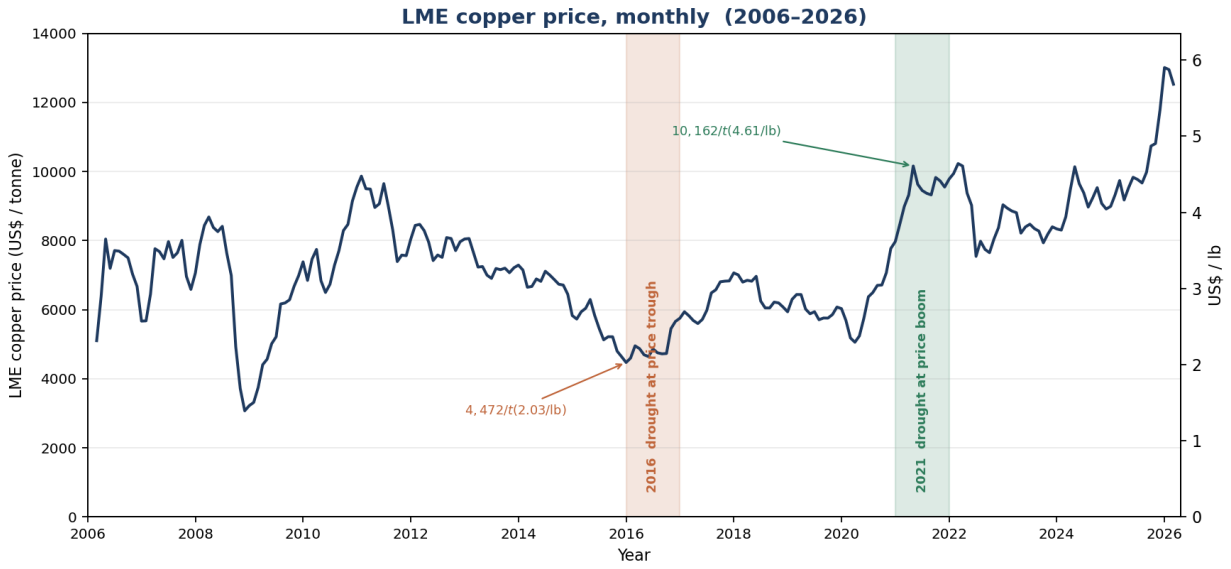


Figure B.9. Monthly copper price trend, 2006–2026.

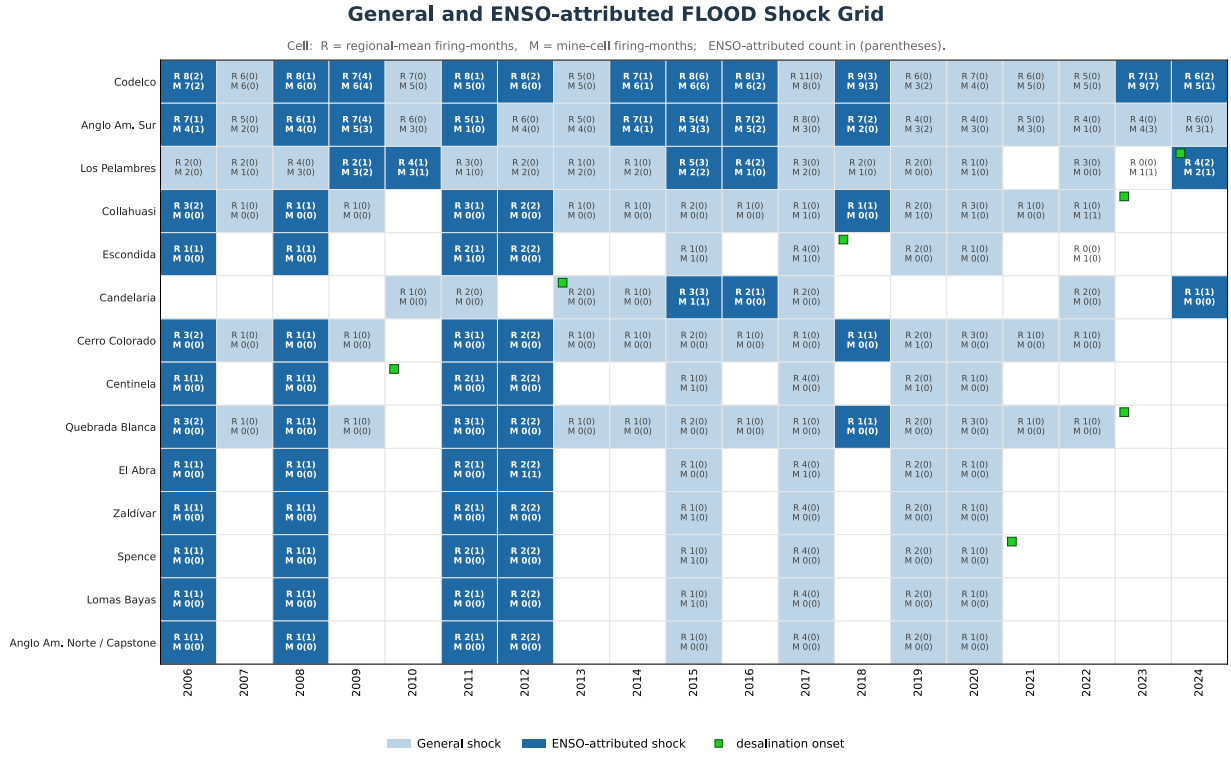


Figure B.10. Flood shock grid, 2006–2024. Each cell reports the general count with the ENSO-attributed subset in parentheses, for two channels: R = regional-mean monthly firing-months, M = mine-cell daily firing-months. Light cells recorded a general flood shock only; dark cells recorded an ENSO-attributed flood shock. Row labels in teal denote desalinated operations (filled square marks desalination onset); black denotes continental water.